

Psychology Statistics Practice Materials with Jamovi, Volume 2

*Categorical, Rank-Based, Quasi-Experimental, and Single-Case
Evidence*

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Problems and Worked Solutions - Version 1.0 - 3 September 2026

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Source, Feedback, and Suggested Attribution

Repository, source files, synthetic datasets, dictionaries, verified result records, and version history:
<https://github.com/nicholaskarlson/data>.

Questions, corrections, teaching ideas, and suggested improvements are welcome in GitHub Discussions.

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Relationship to Volume 1

Volume 1, *Psychology Statistics Practice Materials with Jamovi: Model Choice, Worked Solutions, and Scientific Evidence*, contains Units 1 through 11 and 232 problems with worked solutions. It is available under the same license at the same repository, with version DOI <https://doi.org/10.5281/zenodo.22262048>.

This volume continues that sequence. Its units are numbered 12 through 15 and its problems are numbered accordingly, so a reader may work through both volumes as one continuous resource. Volume 2 does not repeat Volume 1's material and assumes its habits rather than its page numbers.

Contents

This open resource contains four units organized in two parts. Problems and worked solutions are presented together so readers can commit to an answer before consulting the feedback.

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Introduction: When the Familiar Model Does Not Fit

Four situations that a first course reaches and a first course rarely finishes

Volume 1 ended by comparing three or more conditions. That is where many first courses stop teaching model choice and start teaching exceptions. The exceptions are treated as a short chapter near the back of the book, usually titled *nonparametric methods* or *other designs*, and they are usually taught as a list of substitutes: use this test when the assumption fails, use that test when the outcome is a category.

That framing is the problem this volume exists to correct.

None of the four situations in these units is a substitution. Each one changes the question, not only the procedure.

A categorical outcome is not a damaged continuous outcome. When a participant chooses one of three support resources, there is no mean to estimate. The quantity of interest is a pattern of counts, and the analysis compares an observed pattern with the pattern that independence would produce. Calling a chi-square test “the nonparametric alternative” hides the fact that the estimand changed.

A rank-based comparison is not a repaired mean comparison. When a distribution is skewed, a rank method does not compute a more robust mean. It answers a question about the ordering of observations across groups. That question is often the better one, but it is a different one, and its result cannot be reported in the units of the original scale without an argument that nothing in the test supplies.

A quasi-experimental design does not become an experiment because a covariate was added. Adjustment changes which comparison the coefficient describes. Whether that change reduces bias or introduces it depends on what the covariate is - and in particular on whether it was caused by the grouping variable, in which case adjusting for it removes part of the very effect being estimated.

A single-case design does not have a small sample. It has a different unit. Forty-two daily observations of one person are forty-two measurements of one case. The comparison is within a series, and its credibility comes from replication across cases and from the pattern of level, trend, and overlap - not from a p value computed as if the days were people.

Each unit therefore begins one step before the procedure. It asks what the outcome is, what one row represents, what quantity the analysis targets, and what the design permits a reader to conclude. The procedure follows those answers.

The claim ceiling does not move

Volume 1 introduced the **claim ceiling**: the strongest conclusion the complete evidence record authorizes. Nothing in this volume raises it. A significant chi-square does not make an observed association causal. A rank test does not make a convenience sample representative. An analysis of covariance does not manufacture random assignment. A large nonoverlap statistic does not establish that a routine will work for anyone else.

Three of these four designs are especially prone to a particular failure: the analysis is genuinely more sophisticated than a t test, and the added sophistication is quietly transferred to the conclusion. It should not be. Model complexity and identification are independent. A more careful calculation on a design that cannot separate explanations produces a more careful answer to the same limited question.

What each unit uses

Every problem in this volume rests on a public synthetic dataset with a published data dictionary and an independently recomputed verified result record. Nothing here requires a purchase, a login, or a prepared jamovi session file.

Unit	Primary dataset	Central methods
12	study_08_help_seeking_categorical	Frequency and contingency tables, chi-square test of independence, expected counts, Cramér's V
13	study_09_skewed_wellbeing_nonparametric	Distribution inspection, Kruskal-Wallis with tie correction, jamovi epsilon-squared, bias-adjusted rank effect size, Mann-Whitney comparisons with Holm adjustment
14	study_10_developmental_emotion_recognition	Grouped descriptives, one-way ANOVA, Bonferroni comparisons, covariate adjustment, quasi-experimental limits
15	study_11_single_case_habit_tracking	Phase descriptives, ordinal frequencies, visual level and trend inspection, nonoverlap of all pairs

Every numerical value printed in the worked solutions of this volume is recomputed from the source CSV files by a public audit script in the same repository. If a number here disagrees with your jamovi output, one of three things is true: your import differs, your options differ, or there is an error worth reporting. All three are worth finding out.

How to Use These Materials

Read the introduction first. This section describes the practice routine, which is the same routine used in Volume 1.

Statistics becomes usable through decisions, not recognition alone. Reading a clear explanation can make a method feel familiar; solving a new case reveals whether the reasoning can be retrieved and transferred.

Each unit moves through six levels. Begin without looking at the solutions. Commit to an answer, mark your confidence, and then compare your reasoning with the worked response. When an answer is wrong, record the cue you missed rather than merely copying the correction.

The six levels

1. **Recall and recognize:** retrieve the vocabulary and design features.
2. **Predict before clicking:** state what pattern you expect before seeing software output.
3. **Calculate and explain:** connect a small calculation to the model's logic.
4. **Read jamovi output:** locate, translate, and integrate the relevant evidence.
5. **Choose the model:** select one analysis and reject plausible alternatives.
6. **Report and critique:** communicate the result and repair reasoning that exceeds the evidence.

Each unit contains four problems at each level, twenty-four problems in total.

A useful practice loop

For each problem, write four brief notes: **answer**, **reason**, **confidence**, and **revision**. Reattempt missed problems after a delay. A correct answer that depends on remembering the page is less valuable than a correct answer reached from the design and data structure.

When a solution includes **Study in the Larger Evidence Program**, record what the study adds, what it cannot settle, and which next study would be most informative. This keeps one statistical result in its scientific context.

Working in jamovi Desktop

Every analysis in this volume can be reproduced in the free jamovi Desktop application on Windows, macOS, or Linux. The units name analysis goals and output fields rather than reproducing pictures of the interface, because interfaces move between releases and platforms while the statistical goal does not.

Begin every exercise from a fresh import of the named CSV. Before running anything, confirm four things: the row count matches the dictionary, each variable carries the intended measurement level, blank cells are understood as missing rather than as zero, and the group order matches the direction stated in the problem. Most disagreements between a reader's output and a verified record are caused by one of those four items.

Companion data and feedback

The public companion files used by this volume are `study_08_help_seeking_categorical.csv`, `study_09_skewed_wellbeing_nonparametric.csv`, `study_10_developmental_emotion_recognition.csv`, and `study_11_single_case_habit_tracking.csv`, together with their data dictionaries and verified result records. Browse them at <https://github.com/nicholaskarlson/data>; use [Discussions](#) for questions and improvements.

Teaching variants

Several problems refer to a **teaching variant**, written as Study 08A, Study 08B, Study 09A, Study 10A, or Study 11A. Every variant in this volume is a filter or subset of the same public CSV, described exactly in the problem, so a reader can reproduce it in jamovi with a row filter and no new file. Variants exist to prevent model choice from depending on whether a particular result turned out to be favorable.

Optional Books for Deeper Study

This open resource is complete on its own. The following published books are optional purchases for readers who want fuller conceptual instruction, reproducible code workflows, or guidance for planning real research and consulting relationships.

Psychology Research Methods and Statistics by Design with Jamovi

This 27-chapter, beginner-friendly guide is the closest conceptual companion to these practice materials. Chapters 22 through 25 cover the four topics in this volume directly: categorical data and chi-square tests, nonparametric methods and robust thinking, quasi-experimental and developmental designs, and single-case and small-*N* designs. It uses the same public synthetic-data ecosystem. [View on Amazon](#).

Psychological Statistics by Design

This advanced, code-first psychology guide follows evidence from a research question and data layout through Python analysis, independent R verification, parity checks, and reviewer packets. Its chapters on baseline adjustment and analysis of covariance, and on outcomes whose distribution changes the question, extend Units 14 and 13 respectively. [View on Amazon](#).

Applied Statistics with Python and R

This accessible Python-first guide develops a broader applied workflow, using R for optional independent verification and PyStatsV1 as a bridge to a proof-first evidence package. Use it when you want to turn the reasoning practiced here into a reproducible, cross-language workflow. [View on Amazon](#).

Before You Hire a Statistician

This guide explains what statistical consulting can and cannot do, what to prepare before contact, how scope, fees, timelines, deliverables, and revisions work, and how privacy, authorship, ethics, and responsibility should be handled. Use it before adapting these exercises to a real study, sharing data, or hiring statistical help. [View on Amazon](#).

The Reader's Evidence Checklist

Before accepting a statistical conclusion, ask:

1. What is the scientific question?
2. What are the cases or independent units?
3. How were the constructs measured, and are the scores comparable?
4. How were observations sampled, assigned, exposed, and retained?
5. What estimand does the analysis target?
6. Why does the selected model match the design and dependence structure?
7. What estimate and interval answer the question?
8. Which assumptions and diagnostic evidence matter for that answer?
9. What alternative explanations remain?
10. What is the claim ceiling?
11. What does the study contribute to responsiveness, consistency, or mechanism?
12. What next evidence would most reduce the remaining uncertainty?

Use this checklist before opening jamovi, while reading its output, and again when writing the conclusion. The purpose is not to make every claim timid. It is to make each claim as strong as the evidence permits - and no stronger.

Two additions for this volume

Four situations in this volume introduce two further questions that Volume 1 did not need.

13. Does the summary I am about to report exist on the scale I am about to report it on? A chi-square test produces no mean difference. A rank test produces no difference in the outcome's units. Reporting one anyway is the most common error in these four units.

14. What is the unit that could be replicated? In a categorical study it is the participant. In a quasi-experimental comparison it is the group whose selection must be explained. In a single-case design it is the case, and one case cannot support generality no matter how clean its series looks.

PART IV

Outcomes that are not means

Unit 12: Categorical Outcomes and Association

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Unit Purpose

When the outcome is a category rather than a quantity, there is no mean to estimate and no difference to report on the outcome's scale. The analysis compares an observed pattern of counts with the pattern that independence would produce, and the effect size describes the strength of that departure. This unit develops the habit of identifying the case, the outcome's measurement level, the direction of conditioning, and the quantity a categorical analysis can and cannot deliver.

Learning Objectives

After completing this unit, a learner should be able to:

- recognize when an outcome is categorical and state why a mean comparison is unavailable rather than merely inadvisable;
- compute an expected count from row, column, and grand totals and explain what independence means in that formula;
- distinguish row percentages, column percentages, and total percentages, and choose the one that answers the question;
- identify which cells drive a chi-square statistic and describe the pattern in words;
- separate the test statistic, which depends on sample size, from the effect size, which does not;
- state the expected-count condition and what to do when it fails; and
- keep a causal claim inside what the assignment mechanism supports, even when the analysis is correct.

Decision Map

Ask these questions in order:

1. What is the scientific question, and is the outcome a quantity or a category?
2. What are the cases, and does each case contribute exactly one observation to the table?
3. Which variable is the grouping variable and which is the outcome, and does the question ask about the distribution of the outcome within groups?
4. What pattern of counts would independence produce, and are the expected counts large enough to trust the reference distribution?
5. What does the observed departure look like, cell by cell, in words?
6. What effect size describes the strength of association, separately from the sample size?
7. What does the assignment mechanism identify, what is the claim ceiling, and what evidence should come next?

Study Files

The primary dataset is study_08_help_seeking_categorical.csv. Two teaching variants are subsets of the same file, reproducible in jamovi with a row filter. Public CSV files, dictionaries, and verified result records are available at <https://github.com/nicholaskarlson/data>.

Study	Data structure	Primary outcome	Unit role
Study 08	150 participants, 50 in each of three message conditions	Selected support resource, three categories	Balanced randomized comparison of a categorical outcome
Study 08A	The 58 participants with prior_support_use = yes	Selected support resource	Smaller subset; a below-five expected count and an inconclusive test
Study 08B	The 92 participants with prior_support_use = no	Selected support resource	Larger subset; same association strength, different conclusion

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each Pause and Decide, stop reading and write an answer, reason, and confidence rating. The explanation that follows is feedback on that commitment.

Plausible Research Scenario: Three Ways of Describing the Same Support Service

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A fictional student services office offers three kinds of support: an advisor meeting, a peer group, and a self-guided module. All three remain available to every student at all times. The office wants to know whether the way the service is described changes which one students choose.

One hundred fifty synthetic students are randomly assigned in equal numbers to see one of three short messages. The autonomy message emphasizes making one's own plan; the belonging message emphasizes that other students use the service; the skills message emphasizes learning a specific technique. Each student then selects exactly one resource. The research question is: **does the distribution of resource choices differ across the three message conditions?**

One student is the independent unit. Each student contributes one row and appears in exactly one cell of a three-by-three table. The outcome, resource_choice, is nominal. There is no order among advisor meeting, peer group, and self-guided module, and no numerical spacing between them.

A fourth variable, prior_support_use, records whether the student had used the service before. It was measured, not assigned.

Pause and Decide

Before reading further, name the case, state the measurement level of the outcome, decide which percentages answer the question, choose the analysis, and state the strongest claim a randomized message assignment could support. Then say what quantity this design cannot produce no matter how large the sample becomes.

Why This Model?

A chi-square test of independence compares the observed table of counts with the counts expected if resource choice were unrelated to message condition. That matches a question about whether one categorical distribution differs across levels of another categorical variable.

A one-way analysis of variance is unavailable, not merely inferior: resource_choice has no numerical values to average. Recoding the three resources as 1, 2, and 3 and comparing means would impose an order and a

spacing that the measurement does not contain, and the resulting number would change if the labels were listed in a different order. A separate two-proportion test for each resource would answer three narrower questions and would inflate the family-wise error rate without ever describing the table as a whole.

Because assignment to message condition was random and balanced, the design supports a causal claim about the effect of message framing on the distribution of choices among the represented students. It does not support a claim about any particular student, about the merits of the resources themselves, or about students at another institution.

What Would a Favorable Result Look Like?

The office would hope to see a pattern in which at least one message shifts choices toward a resource the office considers underused, with an effect size large enough to justify changing the wording. Direction and pattern matter more than whether a threshold is crossed.

An inconclusive table can also be scientifically successful. If the messages were delivered faithfully and the sample is adequate, a small association tells the office that wording is not the lever it hoped for, which redirects effort toward the service itself.

Best Defensible Claim

Statistical evidence: The chi-square analysis estimates whether, and how strongly, the observed distribution of resource choices departs from independence across the three message conditions, and Cramér's V describes the strength of that association on a scale from 0 to 1.

Causal identification: Random assignment to message condition supports a causal claim about the message's effect on the distribution of choices among the represented participants.

Claim ceiling: The office may claim that message framing changed which resource these students chose. It may not claim that any resource is more effective, that a particular student would have chosen differently, or that the pattern would recur in another population or service context.

What Must Be True?

Each student must contribute exactly one observation, and students' choices must not influence one another. The three categories must be exhaustive and mutually exclusive. Expected counts must be large enough for the chi-square reference distribution to be a reasonable approximation. The messages must have been delivered as intended.

Defensibility Procedures

Prespecify the table, the outcome, and the effect size before inspecting the counts. Confirm the assignment mechanism and check that the measured covariate is balanced across conditions, which is a check on the randomization rather than a hypothesis test of interest. Report the full table with observed counts, expected counts, and one clearly labelled set of percentages; report chi-square with its degrees of freedom, the p value, the effect size, and the minimum expected count.

Next evidence priority: If the goal is to increase useful help-seeking rather than to move choices between options, the next study should measure an outcome that reflects benefit, not selection, and should follow students beyond the moment of choice.

Claim Audit

Repair this claim before continuing: "The chi-square test proves that the belonging message is the best message." Your repair should distinguish an association in a table from a ranking of options, and a statistically detectable pattern from a practically useful one.

Plausible Research Scenario: The Same Question in Two Groups of Different Size

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A second analyst asks whether the message effect is present among students who had used the service before and among students who had not. The analyst splits the same file on `prior_support_use` and runs the same three-by-three analysis twice.

Study 08A contains the 58 students who had used the service before. Study 08B contains the 92 who had not. Both are subsets of the same public file and can be reproduced with a row filter.

This split was not planned before the data were inspected, and `prior_support_use` was measured rather than assigned.

Pause and Decide

Before reading further, predict what will happen to the chi-square statistic, the p value, the effect size, and the minimum expected count in each of the two unequal subsets. Then decide whether a difference in conclusions between the two subsets would be evidence that the message works differently for the two kinds of student.

Why This Model?

Within each subset the analysis is the same chi-square test of independence, and it is correctly specified for the same reason as before. What changes is the interpretive status of the comparison *between* subsets.

Comparing two separately computed chi-square tests is not a test of whether the association differs between subsets. A difference in significance is not a significant difference. The question “does the message effect depend on prior use?” is a question about an interaction, and answering it requires a model that includes both variables and their interaction, not two tests read side by side.

What Would a Favorable Result Look Like?

The analyst would hope to see a similar association in both subsets, which would suggest the pattern is not confined to one kind of student. Similar effect sizes with different p values would be an ordinary consequence of different sample sizes and would not be a finding.

Best Defensible Claim

Statistical evidence: Each subset analysis estimates the strength of association between message and choice within that subset.

Causal identification: Message assignment was random within the whole sample, so the message contrast remains interpretable inside each subset. The comparison between subsets is observational, because prior support use was not assigned.

Claim ceiling: The analyst may describe the association within each subset. The analyst may not claim that the effect of the message differs by prior use on the basis of two separate tests, and may not treat a subset chosen after seeing the overall result as a prespecified analysis.

What Must Be True?

The subset must be defined by a variable measured before the outcome. Expected counts within each subset must still support the approximation. The reader must be told that the split was exploratory.

Defensibility Procedures

Report both subset tables in full, including minimum expected counts. State whether the split was prespecified. If the interaction is the question, fit a model that can answer it and say so. Report effect sizes for both subsets alongside the overall analysis so that a reader can see what changed and what did not.

Next evidence priority: A prespecified study powered for the interaction, with prior support use recorded before assignment and treated as a stratifying variable.

Claim Audit

Explain why “the message worked for new users but not for returning users” is indefensible when it rests on one subset with $p = .017$ and another with $p = .095$.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 12.1 — Name the unavailable quantity

Study 08 records which of three resources each of 150 students selected. A reviewer asks for “the mean resource choice by condition.” Explain what is wrong with the request in terms of measurement level, and name the quantity that is available instead.

Problem 12.2 — Identify the case and the cell

How many rows does Study 08 contribute to the three-by-three table, and how many cells does one student occupy? Name the design feature that would break if a student could select two resources.

Problem 12.3 — State what independence means here

In a chi-square test of independence, what would it mean, in the language of this study, for resource_choice to be independent of message_condition? Answer without using the word “significant.”

Problem 12.4 — Degrees of freedom from the table shape

Study 08’s table has three message conditions and three resource categories. State the degrees of freedom and the rule that produces it. Why does the total sample size not appear in that rule?

Level 2: Predict Before Clicking

Problem 12.5 — Predict the expected counts

Every message condition contains exactly 50 students, and 57 students in total chose an advisor meeting. Before running anything, predict the expected count for the autonomy-message and advisor-meeting cell, and state whether the three expected counts in the advisor-meeting column will be equal to one another.

Problem 12.6 — Predict which cells will dominate

The belonging-message row contains 22 peer-group choices and 8 self-guided choices; the skills-message row contains 7 peer-group choices and 20 self-guided choices. Each of those cells has an expected count of 14.0 or 17.0. Predict which two cells will contribute most to the chi-square statistic, and explain why a cell 9 below an expectation of 17.0 can contribute more than a cell 8 above an expectation of 14.0.

Problem 12.7 — Predict the effect of shrinking the sample

Study 08B contains 92 of the 150 students and shows a similar pattern of percentages. Predict what will happen to the chi-square statistic, the p value, and Cramér’s V relative to the full sample, and say which of the three is designed to be insensitive to sample size.

Problem 12.8 — Predict the expected-count problem

Study 08A contains 58 students. Its smallest row total is 17 and its smallest column total is 16. Predict the smallest expected count in that table and state whether it satisfies the usual condition.

Level 3: Calculate and Explain

Problem 12.9 — Compute an expected count

Using row total 50, column total 42, and grand total 150, compute the expected count for the belonging-message and peer-group cell. Show the formula and explain in one sentence why it is the count implied by independence.

Problem 12.10 — Compute two cell contributions

The belonging-message and peer-group cell has an observed count of 22 against an expected count of 14.0. The belonging-message and self-guided cell has an observed count of 8 against an expected count of 17.0. Compute each cell's contribution to the chi-square statistic and state which is larger.

Problem 12.11 — Recover the effect size

Study 08's chi-square statistic is 17.765 with a grand total of 150 and a table that is three by three. Compute Cramér's V using $V = \sqrt{(\chi^2 / (N \times (k - 1)))}$, where k is the smaller of the number of rows and columns. Explain what the denominator is doing.

Problem 12.12 — Convert a count to the right percentage

In the autonomy-message row, 23 of 50 students chose the self-guided module, and 51 students chose the self-guided module overall. Compute both the row percentage and the column percentage for that cell, and state which one answers the study's research question.

Level 4: Read jamovi Output

Problem 12.13 — Read the contingency table

The verified contingency table for Study 08 is reproduced below, with observed counts and expected counts in parentheses.

Message condition	Advisor meeting	Peer group	Self-guided module	Row total
Autonomy	14 (19.0)	13 (14.0)	23 (17.0)	50
Belonging	20 (19.0)	22 (14.0)	8 (17.0)	50
Skills	23 (19.0)	7 (14.0)	20 (17.0)	50
Column total	57	42	51	150

Describe the pattern in words, naming the resource each message pulled toward and the resource each message pulled away from. Do not use the words "significant" or "caused."

Problem 12.14 — Read the test line

jamovi reports $\chi^2 = 17.765$, $df = 4$, $p = .001$, Cramér's $V = 0.243$, with a minimum expected count of 14.0 and no cells below five. Write one sentence reporting the test in APA style and one sentence saying what the effect size adds that the p value does not.

Problem 12.15 — Read the smaller subset

For Study 08A, jamovi reports $\chi^2 = 7.912$, $df = 4$, $p = .095$, Cramér's $V = 0.261$, minimum expected count 4.69. For Study 08B it reports $\chi^2 = 12.105$, $df = 4$, $p = .017$, Cramér's $V = 0.256$, minimum expected count 7.00. Read the two subset effect sizes together with the full-sample V of 0.243 and state what the comparison shows.

Problem 12.16 — Read a balance check

The same file reports a message-condition by prior-support-use table with $\chi^2 = 1.068$, $df = 2$, $p = .586$, Cramér's $V = 0.084$. Explain what this table is being used for, and why a large p value here is reassuring while a large p value in the main analysis would not be.

Level 5: Choose the Model

Problem 12.17 — Choose the analysis for the stated question

The question is whether the distribution of resource choices differs across the three message conditions, with one categorical choice per student. Choose the analysis and justify it from the data structure. Then reject a one-way analysis of variance on numerically recoded resource choices, and state the specific way that recoding changes the answer.

Problem 12.18 — Reject the three separate tests

An analyst proposes running three separate two-proportion tests, one per resource, and reporting whichever is significant. Give two distinct reasons this is indefensible, one about the family of tests and one about what question the table was built to answer.

Problem 12.19 — Reject the subset conclusion

An analyst reports that the message “worked” in Study 08B ($p = .017$) but “did not work” in Study 08A ($p = .095$), and concludes that the effect depends on prior support use. Reject that conclusion. Name the analysis that would actually address the question and the design feature that limits it.

Problem 12.20 — Choose a response to a sparse cell

Study 08A has a minimum expected count of 4.69, just below the usual threshold. List three defensible responses, and one indefensible one. Explain why deleting the smallest category is the indefensible option.

Level 6: Report and Critique**Problem 12.21 — Write the result paragraph**

Using the verified values for Study 08, write a short APA-style results paragraph. Include the table shape, the test statistic with degrees of freedom, the p value, the effect size, a one-clause description of the pattern, and the expected-count check.

Problem 12.22 — Repair a causal overreach

Repair this sentence: “Because students who saw the belonging message were more likely to choose the peer group, peer support is what students actually need.” Your repair should preserve what the design does license.

Problem 12.23 — Repair a scale error

Repair this sentence: “The belonging message increased peer-group selection by 8 students on average, $\chi^2(4) = 17.77$, $p = .001$.” Identify every distinct error. The arithmetic in the sentence is correct.

Problem 12.24 — Write the limitation that keeps the claim honest

Study 08 is randomized, correctly analysed, and clearly reported. Write the two-sentence limitation that should still accompany it. One sentence must concern what the outcome measures; the other must concern to whom the result applies.

Solutions

Level 1 Solutions

Solution 12.1

resource_choice is nominal. Its values are labels for three distinct options with no order and no numerical spacing, so no arithmetic operation on them is meaningful and no mean exists to estimate. The available quantities are counts and the proportions or percentages derived from them: within each message condition, the proportion of students choosing each resource. Decisive cue: ask whether the halfway point between two values is interpretable. There is no resource halfway between an advisor meeting and a peer group.

Solution 12.2

Study 08 contributes 150 rows, one per student, and each student occupies exactly one of the nine cells, so the nine cell counts sum to 150. If a student could select two resources, the counts would exceed the number of students and the cells would no longer be mutually exclusive. The chi-square test of independence assumes each case appears once; a multiple-response question requires a different analysis.

Solution 12.3

Independence would mean that knowing which message a student saw tells you nothing about which resource they chose: the percentage choosing an advisor meeting, a peer group, and a self-guided module would be the same in all three message rows, apart from sampling variation. Equivalently, each cell count would be close to the count implied by multiplying its row proportion by its column proportion and by the grand total.

Solution 12.4

$df = (\text{rows} - 1) \times (\text{columns} - 1) = (3 - 1) \times (3 - 1) = 4$. The rule counts how many cells are free to vary once the row and column totals are fixed: fill in a two-by-two corner of the table and every remaining cell is determined by subtraction. Sample size does not appear because degrees of freedom describe the shape of the table, not how much information it contains. Sample size enters the analysis elsewhere, through the expected counts and therefore through the size of the statistic.

Level 2 Solutions

Solution 12.5

Expected count = $50 \times 57 / 150 = 19.0$. The three expected counts in the advisor-meeting column are all 19.0, because all three row totals are equal at 50. Equal expected counts within a column are a consequence of the balanced design, not of the result. Transfer cue: expected counts are built from the margins alone and can be written down before any cell is inspected.

Solution 12.6

The two dominant cells are belonging with peer group (22 observed against 14.0 expected, a deviation of +8) and belonging with self-guided (8 observed against 17.0 expected, a deviation of -9). The second contributes more, even though its expected count is larger, because each contribution is a squared deviation divided by its own expected count, so both the size of the deviation and the size of the expectation matter. Here $9^2 / 17.0 = 4.765$ exceeds $8^2 / 14.0 = 4.571$: squaring the deviation outweighs the larger divisor.

Solution 12.7

With a similar pattern of percentages and roughly two thirds of the cases, the chi-square statistic falls and the p value rises. Cramér's V should stay close to its full-sample value, because it is a standardized measure of association that divides the statistic by the sample size. The verified values bear this out: χ^2 falls from 17.765 to 12.105 and p rises from .001 to .017, while V moves only from 0.243 to 0.256. The effect size is the quantity designed to be insensitive to N .

Solution 12.8

The smallest expected count is the product of the smallest row total and the smallest column total divided by the grand total: $17 \times 16 / 58 = 4.69$. That falls below the customary minimum of five, so the chi-square approximation is at its limit and the analysis should say so. The condition concerns expected counts, not observed ones; a cell may contain few observations and still be acceptable if its expectation is adequate.

Level 3 Solutions

Solution 12.9

Expected count = row total $\times$ column total / grand total = $50 \times 42 / 150 = 14.0$. Under independence, the proportion choosing a peer group should be the same in every row, so the belonging row's share of the 42 peer-group choices should be its share of the sample, $50 / 150$, giving $42 \times (50 / 150) = 14.0$. The formula is that statement rearranged.

Solution 12.10

Peer group: $(22 - 14.0)^2 / 14.0 = 64 / 14.0 = 4.571$. Self-guided: $(8 - 17.0)^2 / 17.0 = 81 / 17.0 = 4.765$. The self-guided cell contributes slightly more because its deviation is larger, even though its larger expected count divides that deviation more heavily. Together these two cells contribute 9.336 of the 17.765 total, which is why the belonging row is the clearest part of the pattern.

Solution 12.11

$V = \sqrt{(17.765 / (150 \times (3 - 1)))} = \sqrt{(17.765 / 300)} = \sqrt{0.05922} = 0.243$. The denominator is the largest value the chi-square statistic could take in a table of this size and sample: N times one less than the smaller table dimension. Dividing by it rescales the statistic onto a 0-to-1 range so that association strength can be compared across tables with different sample sizes and shapes.

Solution 12.12

Row percentage = $23 / 50 = 46.0$ percent of autonomy-message students chose the self-guided module. Column percentage = $23 / 51 = 45.098$ percent of self-guided choosers had seen the autonomy message. The research question asks how choices are distributed *within* each message condition, so the row percentage answers it. The column percentage conditions in the wrong direction; it describes the composition of a group defined by the outcome, which is not what a randomized message assignment is designed to inform.

Level 4 Solutions

Solution 12.13

Students who saw the autonomy message chose the self-guided module more often than independence would imply (23 against 17.0 expected) and chose an advisor meeting less often (14 against 19.0). Students who saw the belonging message chose the peer group markedly more often (22 against 14.0) and the self-guided module markedly less often (8 against 17.0). Students who saw the skills message showed the opposite peer-group pattern (7 against 14.0) and chose an advisor meeting somewhat more often (23 against 19.0). Each message shifted choices toward the resource that most closely matched its framing, and the belonging and skills rows moved in opposite directions on the peer-group option.

Study in the Larger Evidence Program: Message Framing and Resource Choice

- **What this study adds:** A randomized estimate of how message framing redistributes choices among three concurrently available resources, with a full table and a standardized association measure.
- **What it cannot settle:** Whether any of the three resources helps, whether a shifted choice improves an outcome the student cares about, or whether the pattern recurs in another service or population.
- **Causal-support role:** Responsiveness evidence for the framing manipulation, because the message was deliberately assigned. It supplies no mechanism evidence and no evidence about benefit.

- **Next evidence priority:** Follow the same randomized framing through to a downstream outcome such as attendance or self-reported benefit, so that the study estimates a consequence rather than a selection.

Solution 12.14

Resource choice was associated with message condition, $\chi^2(4, N = 150) = 17.77, p = .001$, Cramér's $V = 0.243$; no expected count fell below five, with a minimum of 14.0. The effect size adds the strength of the association on a scale that does not depend on the sample size: a V of 0.243 describes a modest but not negligible departure from independence, whereas the p value only reports how surprising the table would be under an independence model. Two tables with identical percentages and different sample sizes share an effect size and differ in p .

Solution 12.15

The three effect sizes are close together: 0.243 in the full sample, 0.256 in the larger subset, and 0.261 in the smaller subset. The p values are not: .001, .017, and .095. The comparison shows that the strength of association is essentially stable across the three analyses and that the differences in conclusion are produced by sample size, not by a change in the pattern. Reading the p values alone would suggest a finding that appears and disappears; reading the effect sizes shows one pattern estimated with three levels of precision. The Study 08A analysis additionally carries a minimum expected count of 4.69 and should be reported with that qualification.

Solution 12.16

The message-condition by prior-support-use table is a randomization check: prior support use was measured before the outcome and was not assigned, so under correct random assignment it should be distributed similarly across the three message conditions. The observed near-independence ($V = 0.084$) is consistent with assignment having worked. A large p value is reassuring here because the null model is what the design is supposed to produce. In the main analysis the null model is what the study is trying to rule out, so a large p value there is an absence of evidence rather than evidence of balance. In neither case does a large p value prove that the null is exactly true.

Level 5 Solutions

Solution 12.17

A chi-square test of independence on the three-by-three table is the defensible analysis. Each of 150 students contributes one nominal outcome, the grouping variable is nominal with three levels, and the question concerns whether the distribution of the outcome differs across those levels.

A one-way analysis of variance on recoded values is not merely weaker; it answers a question that does not exist. Coding advisor meeting as 1, peer group as 2, and self-guided module as 3 asserts that a peer group is one unit above an advisor meeting and that a self-guided module is exactly as far above a peer group as a peer group is above an advisor meeting. Relabelling the same three categories in a different order produces different means, a different F , and possibly a different conclusion from the identical data. A result that changes when the labels are reordered is not a result.

Solution 12.18

First, three tests conducted at the same nominal level inflate the family-wise error rate, and selecting the significant one afterwards inflates it further; the reported p value would no longer describe the procedure that generated it. Second, the three tests answer three separate questions about individual resources and never address the question the table was built for, which concerns the distribution across all three options simultaneously. A pattern in which one resource gains and another loses is visible in the table and invisible in any single two-proportion test. Prespecified follow-up comparisons after an overall test, with an explicit multiplicity adjustment, are a defensible alternative; choosing among unadjusted tests after seeing them is not.

Solution 12.19

The conclusion compares two significance decisions and treats their difference as a finding. It is not: a difference in significance is not a significant difference. The two subset effect sizes are 0.261 and 0.256, which are nearly identical; only the precision differs, because one subset has 58 cases and the other 92. The question “does the message effect depend on prior support use?” is a question about an interaction and requires a model containing message condition, prior support use, and their interaction. Even that analysis is limited here: prior support use was measured rather than assigned, so any interaction it reveals is an observational moderation, and the subgroup split was chosen after the overall result was seen.

Solution 12.20

Three defensible responses: report the analysis with the minimum expected count stated so the reader can judge the approximation; use an exact test that does not rely on the large-sample approximation; or collapse categories only if a substantively meaningful grouping exists and the decision is prespecified and reported. A fourth reasonable response is to decline to analyse the subset at all and report the full-sample result, saying that the subset was underpowered.

Deleting the smallest category is indefensible because it discards cases on the basis of the outcome, changes the estimand from “the distribution across three resources” to “the distribution across two,” and does so for a reason that has nothing to do with the research question. It also usually improves the p value, which makes it a decision the reader cannot distinguish from result shopping.

Study in the Larger Evidence Program: The Subgroup Split

- **What this study adds:** A demonstration that the same association, estimated at three sample sizes, yields three different threshold verdicts while the effect size barely moves ($V = 0.243, 0.256, 0.261$).
- **What it cannot settle:** Whether the message effect genuinely differs between students with and without prior support use. Two separate tests cannot answer that question, and the split was not prespecified.
- **Causal-support role:** None on its own. The message contrast retains its randomized status inside each subset, but the subset contrast is observational.
- **Next evidence priority:** A study powered for the interaction, with prior support use recorded before assignment and used as a stratifying variable rather than as a post hoc split.

Level 6 Solutions**Solution 12.21**

One hundred fifty synthetic participants were randomly assigned in equal numbers to one of three message conditions and each selected one of three support resources. Resource choice was associated with message condition, $\chi^2(4, N = 150) = 17.77, p = .001$, Cramér’s $V = 0.243$. The clearest departures from independence occurred in the belonging-message row, where 22 of 50 students chose the peer group against 14.0 expected and 8 chose the self-guided module against 17.0 expected. All expected counts exceeded five, with a minimum of 14.0. Because message condition was randomly assigned, the analysis supports a causal claim about the effect of message framing on the distribution of choices among the represented participants; it does not evaluate the resources themselves.

Solution 12.22

Repaired: “Students who saw the belonging message chose the peer group more often than independence would imply, which indicates that the framing of the description influenced selection among the represented participants. The study measured which resource students chose, not whether any resource helped them, so it cannot establish what students need.”

The original sentence makes two jumps. It moves from a shift in selection to a claim about benefit, which the outcome never measured, and it moves from an effect of the *message* to a property of the *resource*, when the randomized manipulation was the wording and not the service.

Solution 12.23

Three distinct errors. First, a scale error: a chi-square analysis produces no average and no per-student increase, so “increased by 8 students on average” reports a quantity the model never estimated. The 8 is a raw count in a single cell, and the 22-against-14.0 comparison is a departure from an expectation, not a difference in a mean. Second, a granularity error: the chi-square statistic and its p value describe the whole table, so attaching them to one cell’s pattern implies that a specific cell was tested when it was not. Third, a reporting error: the sentence gives no effect size, no expected-count information, and no table shape, leaving the reader unable to judge either strength or adequacy.

A defensible version reports the cell pattern descriptively and the test at the level it was computed: “Among students who saw the belonging message, 22 of 50 (44.0 percent) chose the peer group, compared with 14.0 expected under independence; across the full table, $\chi^2(4, N = 150) = 17.77, p = .001$, Cramér’s $V = 0.243$.”

Solution 12.24

“The outcome recorded which resource a student selected immediately after reading one message; it did not record whether the selected resource was attended, completed, or beneficial, so the study estimates a change in choice rather than a change in outcome. The participants are a synthetic sample representing one service context, and the result does not establish that the same framing would redistribute choices in another institution, with a different set of available resources, or at a different point in an academic term.”

Both sentences are necessary and they do different jobs. The first bounds the construct: a well-executed randomized study still only measures what the outcome variable measured. The second bounds the population: random assignment supports internal comparison and says nothing about external reach.

Unit 13: Rank-Based Comparisons and Robust Thinking

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Unit Purpose

A rank-based comparison is not a repaired mean comparison. It replaces the observed values with their positions in the combined ordering and asks whether observations from one group tend to sit higher in that ordering than observations from another. The result is a statement about ordering, not about the outcome's scale, and it cannot be reported in the outcome's units without an assumption the test never supplies. This unit develops the habit of choosing a rank method for a stated reason, reporting it on the scale it actually estimates, and refusing to treat outlier deletion as a neutral act.

Learning Objectives

After completing this unit, a learner should be able to:

- describe a distribution from a five-number summary and say what a mean and a standard deviation conceal;
- explain what replacing values with ranks preserves and what it discards;
- compute a mean rank and check rank sums against their known total;
- state what the Kruskal-Wallis test compares and why it is not a test of medians in general;
- interpret jamovi epsilon-squared, a bias-adjusted rank effect size, and a rank-biserial correlation on the scales they occupy;
- apply and justify a multiplicity adjustment for planned pairwise comparisons;
- recognize that a grouping variable can be ordinal while its pattern is not monotonic; and
- treat the deletion of flagged observations as a modelling decision that must be prespecified and reported.

Decision Map

Ask these questions in order:

1. What is the scientific question, and does it concern typical values, ordering, or the whole distribution?
2. What are the cases, and does each contribute one independent observation?
3. What does the outcome's distribution look like within each group, before any test is chosen?
4. What is the stated reason for a rank-based method: skew, outliers, unequal spread, an ordinal outcome, or something else?
5. What quantity does the chosen method estimate, and on what scale can it be reported?
6. If the overall test is followed by pairwise comparisons, what family are they in and how is multiplicity handled?
7. What does the design identify, what is the claim ceiling, and what evidence should come next?

Study Files

The primary dataset is `study_09_skewed_wellbeing_nonparametric.csv`. The teaching variant is a subset of the same file, reproducible in jamovi with a row filter on `outlier_flag`. Public CSV files, dictionaries, and verified result records are available at <https://github.com/nicholaskarlson/data>.

Study	Data structure	Primary outcome	Unit role
Study 09	132 participants, 44 in each of three app-use groups	Wellbeing score, skewed, with three flagged observations	Complete rank-based comparison
Study 09A	The 129 rows with <code>outlier_flag = no</code>	Wellbeing score	The consequence of deleting flagged observations

The variable `outlier_flag` is a teaching flag included in the synthetic file to mark the observations an inattentive analyst is most likely to delete. It is not a statistical verdict and it does not identify errors.

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each Pause and Decide, stop reading and write an answer, reason, and confidence rating.

Plausible Research Scenario: Wellbeing Across Three Levels of App Use

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A fictional student wellbeing service surveys 132 students. Each student reports how much they use a wellbeing app - low, moderate, or high - and completes a wellbeing questionnaire scored from roughly 0 to 100. The groups are formed by what students report, not by anything the service assigned.

The wellbeing scores are not symmetric. The low-use group in particular has a long lower tail: its minimum is 26.9 against a first quartile of 51.8, and two of its observations carry the teaching outlier flag. A third flagged observation sits in the moderate group.

The research question is: **do wellbeing scores tend to differ across the three app-use groups?** The word *tend* is deliberate. The team is asking whether students in one group generally sit higher in the ordering of wellbeing scores than students in another, not whether one group's arithmetic mean exceeds another's by a stated number of points.

`app_use_group` is recorded as ordinal, with low below moderate below high. Nothing in the design guarantees that wellbeing changes monotonically across those three levels.

Pause and Decide

Before reading further, write down what you expect the distributions to look like from the five-number summaries alone, choose a comparison method and state the reason in one sentence, and predict whether the pattern across the three ordered groups will be monotonic. Then state what quantity your chosen method will and will not produce.

Why This Model?

The Kruskal-Wallis test compares the mean ranks of three or more independent groups after replacing all observations with their positions in the combined ordering. That matches a question about whether observations from one group tend to sit higher than observations from another, and it is not disturbed by a long tail in the way that a mean is.

The reason to prefer it here is skew and a small number of extreme low observations, not unequal spread. The three groups' standard deviations are 9.461, 7.167, and 7.479, and a median-centred Brown-Forsythe test gives

$F(2, 129) = 0.54, p = .583$. Spread is not the problem. Saying so matters, because “the variances were unequal” is the most commonly given and most commonly wrong justification for a rank method.

A one-way analysis of variance is available and would not be absurd. It gives $F(2, 129) = 24.77, p < .001$, with eta-squared of 0.278, and it points to the same ordering. The choice between the two is therefore not a choice between a significant and a non-significant result. It is a choice about which quantity to report: a comparison of arithmetic means, which the long lower tail makes a less representative summary of a typical student, or a comparison of orderings, which is insensitive to how far the tail extends.

The Kruskal-Wallis test is often described as a test of medians. It is not, in general. It is a test of whether one group’s observations tend to be larger, and it becomes a test about medians only under the additional assumption that the groups’ distributions have the same shape and differ only by a shift.

What Would a Favorable Result Look Like?

The service would hope to see a clear and interpretable ordering across the three groups, with an effect size large enough to justify further work and pairwise comparisons that locate where the differences sit. A non-monotonic pattern would be more interesting than a monotonic one, because it would rule out the simplest story.

An inconclusive result would still be useful. If wellbeing does not order with reported app use, the service learns that self-reported usage volume is not a promising screening variable.

Best Defensible Claim

Statistical evidence: The rank-based analysis estimates whether observations in one app-use group tend to sit higher in the combined ordering of wellbeing scores than observations in another. jamovi’s epsilon-squared, computed as $H / (N - 1)$, summarizes the omnibus separation on the rank scale; the separately calculated bias-adjusted rank effect size subtracts the null expectation from H before rescaling it.

Causal identification: Group membership was reported, not assigned. Nothing in the design separates the effect of app use from the many reasons a student’s wellbeing and their app use might move together, including reverse causation: a student whose wellbeing is poor may use the app more.

Claim ceiling: The service may describe an observed ordering of wellbeing across self-reported usage groups in this sample. It may not claim that app use changes wellbeing, that reducing use would raise wellbeing, or that the pattern would appear in another sample.

What Must Be True?

Each student must contribute one independent observation. The outcome must be at least ordinal, which it is. If the result is to be described as a difference in medians rather than in ordering, the three distributions must have similar shapes - a claim that must be inspected, not assumed.

Defensibility Procedures

Inspect the distributions before choosing the method and record what was seen. Prespecify the method, the effect size, the pairwise comparisons, and the multiplicity adjustment. Report group sizes, five-number summaries, the overall test with its degrees of freedom and effect size, and every pairwise comparison in the family - not only the ones that reached a threshold. State the tie handling and the direction convention for the effect size.

Next evidence priority: A design that measures app use objectively and follows students over time, so that the temporal order of change can be observed rather than assumed.

Claim Audit

Repair this claim before continuing: “High app use lowers wellbeing by about 12 points, $H = 39.57, p < .001$.” Your repair should identify a scale error, a causal error, and a missing effect size.

Plausible Research Scenario: The Analyst Who Cleaned the Data

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A second analyst opens the same file, notices the outlier_flag column, and deletes the three flagged rows before analysing. Two came from the low-use group and one from the moderate-use group. The high-use group is unchanged. The remaining file, Study 09A, has 129 rows with group sizes of 42, 43, and 44.

The analyst reports the cleaned analysis and does not mention the deletion.

Pause and Decide

Before reading further, predict what deleting three low observations from two of three groups will do to the low group's mean, its standard deviation, the overall test statistic, and the effect size. Say explicitly which direction you expect each to move, and then commit to whether you think "cleaning" makes a result more conservative.

Why This Model?

The analysis run on Study 09A is the same Kruskal-Wallis test, correctly specified for the reduced file. That is exactly why it is dangerous: nothing in the output announces that the file changed.

The deletion is a modelling decision, not a data-quality repair. The flagged values are inside the plausible range of the instrument, and no measurement error was identified. Removing them changes the estimand from "wellbeing scores among the students surveyed" to "wellbeing scores among the students surveyed excluding those with unusually low scores" - a population defined partly by the outcome.

What Would a Favorable Result Look Like?

There is no favorable result here, only an honest one. If deletion is genuinely justified - a documented measurement failure, an ineligible participant, a value outside the instrument's possible range - the justification exists before the analysis and is reported. If it does not, the defensible move is to run the analysis both ways and report both.

Best Defensible Claim

Statistical evidence: The cleaned analysis estimates the same rank-based quantity on a subset of the sample defined by the outcome variable.

Causal identification: Unchanged and still absent; deletion does not improve identification.

Claim ceiling: Lower than the original, not higher. A result computed on cases selected partly by their outcome values supports a narrower claim, and the narrowing must be stated.

What Must Be True?

For deletion to be defensible, the rule must be prespecified, applied without reference to the result, applied identically to every group, and reported. None of those conditions is met by noticing a flag column and acting on it.

Defensibility Procedures

Report the analysis on the complete file as the primary result. Report the deletion as a sensitivity analysis with both statistics visible. State the rule, the number of cases removed, and which groups they came from. Never present a cleaned result as the only result.

Next evidence priority: A prespecified data-handling plan written before collection, naming the exclusion rules and the sensitivity analyses that will accompany them.

Claim Audit

Explain why “we removed three outliers to be conservative” is not a defensible sentence, using the verified statistics from Study 09 and Study 09A.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 13.1 — Name what ranking preserves and discards

Replacing 132 wellbeing scores with their positions in the combined ordering preserves one property of the data and discards another. Name both, and give one consequence of each for interpretation.

Problem 13.2 — State what the overall test compares

Complete the sentence precisely: “The Kruskal-Wallis test asks whether ...” Then state the additional condition that would be required before the result could be described as a comparison of medians.

Problem 13.3 — Identify the design

Students reported their own app-use level. Name the design, state whether random assignment is present, and name one alternative explanation for an association that this design cannot exclude and that runs in the opposite direction from the obvious one.

Problem 13.4 — Recognize the effect-size scale

Study 09 reports jamovi epsilon-squared of 0.302, a bias-adjusted rank effect size of 0.291, and a rank-biserial correlation of 0.750 for one pair. State the range each can take, what value indicates no effect, and what each one summarizes.

Level 2: Predict Before Clicking

Problem 13.5 — Predict from the five-number summaries

The low-use group has minimum 26.9, first quartile 51.8, median 55.65, third quartile 61.675, and maximum 78.0. Predict the shape of this distribution and predict whether its mean will be above or below its median. The verified mean is 56.420. Explain the apparent tension between the long lower whisker and that mean.

Problem 13.6 — Predict the ordering

app_use_group is ordinal, running low, moderate, high. The verified group medians are 55.65, 62.10, and 49.00 respectively. Predict the order of the three mean ranks and say whether a test for a linear trend across the ordered groups would represent this pattern well.

Problem 13.7 — Predict the effect of the tie correction

Among the 132 wellbeing scores there are 110 distinct values; 16 values occur more than once, and the largest group of identical values contains four observations. Predict whether the tie correction will raise or lower the test statistic, and predict roughly how large the adjustment will be.

Problem 13.8 — Predict the consequence of deletion

Two of the three flagged observations are low scores in the low-use group, whose median is already the middle of the three. Predict the direction of change in the low group’s mean rank, in the overall statistic, and in the effect size when those rows are removed. State your prediction before reading Solution 13.8.

Level 3: Calculate and Explain

Problem 13.9 — Check the rank sums

The three rank sums are 3029.0, 3999.5, and 1749.5. Compute their total and check it against the total of the ranks 1 through 132, which is $N(N + 1) / 2$. Explain what this check does and does not verify.

Problem 13.10 — Compute a mean rank

The moderate-use group has a rank sum of 3999.5 and 44 members. Compute its mean rank. Compute the mean rank expected if group membership were unrelated to wellbeing, using $(N + 1) / 2$, and state the size and direction of the departure.

Problem 13.11 — Recover both omnibus rank effect sizes

Study 09 reports $H = 39.573$ with three groups and 132 observations. First use jamovi's definition, $\epsilon^2 = H / (N - 1)$, to compute epsilon-squared. Then use $(H - k + 1) / (N - k)$ to compute the bias-adjusted rank effect size. Explain why the second formula subtracts $k - 1$ from H before dividing.

Problem 13.12 — Recover a rank-biserial correlation

For the moderate-versus-high comparison, $U = 1694.0$ with 44 observations in each group. Using $r = 2U / (n_1 n_2) - 1$, compute the rank-biserial correlation and state, in a sentence a non-statistician could read, what it means about pairs of students.

Level 4: Read jamovi Output

Problem 13.13 — Read the grouped descriptives

Group	<i>n</i>	Mean	Median	SD	Min	Q1	Q3	Max	Flagged
Low	44	56.420	55.65	9.461	26.9	51.800	61.675	78.0	2
Moderate	44	61.834	62.10	7.167	47.9	57.175	66.500	82.3	1
High	44	49.702	49.00	7.479	31.3	44.775	55.325	65.4	0

Describe the three distributions in words. Name the one group whose mean and standard deviation are least representative of its typical member, and give the specific numerical evidence.

Problem 13.14 — Read the overall test

jamovi reports $\chi^2 = 39.573$ with $df = 2$ and $p < .001$ for the Kruskal-Wallis test, with mean ranks of 68.841, 90.898, and 39.761 for the low, moderate, and high groups. Its epsilon-squared is $\epsilon^2 = H / (N - 1) = 0.302$. The separately calculated bias-adjusted rank effect size is $(H - k + 1) / (N - k) = 0.291$. Write one APA-style sentence reporting the test, and one further sentence describing the pattern that the mean ranks show.

Problem 13.15 — Read the pairwise family

Comparison	<i>U</i>	<i>z</i>	<i>p</i>	Holm <i>p</i>	Rank-biserial
Low vs moderate	620.5	-2.896	.00378	.00378	-0.359
Low vs high	1418.5	3.756	.000173	.000346	0.465
Moderate vs high	1694.0	6.055	1.41×10^{-9}	4.22×10^{-9}	0.750

Read the direction of each comparison from the sign of the rank-biserial correlation, given the convention that a positive value favours the first-named group. State which pair is furthest apart and which is closest, and explain why the Holm-adjusted values differ from the unadjusted ones for two comparisons but not for the third.

Problem 13.16 — Read the two analyses side by side

For Study 09, $H = 39.573$, jamovi $\epsilon^2 = 0.302$, and the bias-adjusted rank effect size is 0.291, with low-group mean 56.420 and SD 9.461. For Study 09A, after deleting the three flagged rows, $H = 41.560$, jamovi $\epsilon^2 = 0.325$, and the bias-adjusted rank effect size is 0.314, with low-group mean 57.629 and SD 7.755. State what happened to each quantity and explain in one sentence why this comparison makes the phrase “removed to be conservative” untenable.

Level 5: Choose the Model**Problem 13.17 — Choose the analysis and state the real reason**

Choose the comparison for the stated question and justify it from the inspected distributions. Then explain why “the variances were unequal” would be a false justification in this case, citing the specific evidence.

Problem 13.18 — Reject the median claim

An analyst reports: “A Kruskal-Wallis test showed that the median wellbeing score differed across app-use groups.” Explain the precise sense in which this is wrong, name the assumption that would make it right, and give a defensible replacement sentence.

Problem 13.19 — Reject the trend model

An analyst notes that `app_use_group` is ordinal and proposes replacing the Kruskal-Wallis test with a test for a linear trend across the three ordered levels, arguing that it will have more power. Reject the proposal using the verified mean ranks, and state the circumstance in which the proposal would have been sensible.

Problem 13.20 — Reject the unadjusted pairwise report

An analyst runs the three pairwise Mann-Whitney comparisons, reports the two smallest p values, and omits the third because “it was in the same direction anyway.” Give two distinct reasons this is indefensible, and name what should have been prespecified.

Level 6: Report and Critique**Problem 13.21 — Write the result paragraph**

Using the verified values, write a short APA-style results paragraph for Study 09. Include group sizes, a description of the distributions, the overall test with degrees of freedom and effect size, the pairwise family with its adjustment, and the observational limitation.

Problem 13.22 — Repair a scale error

Repair this sentence: “Students with high app use scored 12.13 points lower than students with moderate app use, $H(2) = 39.57$, $p < .001$, $\epsilon^2 = 0.30$.” The arithmetic in the sentence is correct. Explain what is nevertheless wrong and write a version that reports both quantities honestly.

Problem 13.23 — Repair an undisclosed deletion

An analyst reports the Study 09A results with no mention of the deletion. Write the two sentences that must be added to the methods and the one sentence that must be added to the results, and explain which of the two reported statistics should be the primary one.

Problem 13.24 — Write the limitation

Study 09 is correctly analysed, clearly reported, and observational. Write the three-sentence limitation it requires: one sentence on what the grouping variable is, one on the direction of any causal story, and one on what the rank-based statistics do and do not describe.

Solutions

Level 1 Solutions

Solution 13.1

Ranking preserves the *ordering* of the observations and discards the *distances* between them. The consequence of preserving order is that any monotonic change of scale - adding a constant, multiplying by a positive number, applying a logarithm - leaves the analysis unchanged, so the result does not depend on an arbitrary scoring choice. The consequence of discarding distance is that the analysis cannot report how far apart the groups are on the wellbeing scale. A student 30 points below everyone else and a student 3 points below everyone else occupy the same rank.

Solution 13.2

“The Kruskal-Wallis test asks whether observations drawn from one group tend to occupy higher positions in the combined ordering than observations drawn from another.” Describing the result as a comparison of medians additionally requires that the three distributions have the same shape and differ only by a location shift. Under that assumption the ordering statement and the median statement coincide; without it they can diverge, and two groups can have identical medians while one group’s observations tend to be larger.

Solution 13.3

The design is a cross-sectional observational survey with self-reported group membership and no random assignment. Among the alternative explanations it cannot exclude, the most important is reverse causation: a student whose wellbeing is already low may use a wellbeing app more, so app use would be a consequence of low wellbeing rather than a cause of it. The data contain no temporal information that could separate the two directions.

Solution 13.4

jamovi epsilon-squared runs from 0 to 1, with 0 indicating no rank separation. The bias-adjusted rank effect size runs up to 1 but can be slightly negative when H is smaller than its null expectation; 0 occurs when $H = k - 1$. Both summarize omnibus separation on the rank scale rather than variability in wellbeing points. The rank-biserial correlation runs from -1 to $+1$, with 0 indicating no tendency; it is the difference between the proportion of cross-group pairs favouring the first-named group and the proportion favouring the second. None of the three is expressed in wellbeing points.

Level 2 Solutions

Solution 13.5

The five-number summary shows an asymmetric distribution. The lower whisker is long ($55.65 - 26.9 = 28.75$) while the upper whisker is shorter ($78.0 - 55.65 = 22.35$), which indicates a tail extending toward low scores. Within the box, however, the median sits closer to the first quartile ($55.65 - 51.8 = 3.85$) than to the third ($61.675 - 55.65 = 6.025$), so the central half of the data leans the other way.

That is the resolution of the apparent tension: the mean of 56.420 lies slightly *above* the median of 55.65 because the right-leaning central mass outweighs the two extreme low observations in this sample. A long tail does not determine the sign of the mean-median difference by itself; it determines how unstable that difference is. Transfer cue: a five-number summary describes shape more reliably than any single comparison of mean and median.

Solution 13.6

The mean ranks should order moderate highest, then low, then high - the same order as the medians, because ranking preserves ordering. The verified mean ranks are 90.898, 68.841, and 39.761, confirming it.

A test for a linear trend across low, moderate, high would represent this pattern badly. The pattern rises from low to moderate and then falls sharply to high; a linear trend model would average those two opposite movements into a single slope and could report a weak or absent trend from data showing a strong non-monotonic pattern. Ordinal grouping permits a trend test; it does not justify one.

Solution 13.7

The tie correction divides the uncorrected statistic by a quantity slightly less than 1, so it raises the statistic. With only 16 tied values among 110 distinct values and no tie group larger than four, the correction is very small. The verified correction factor is 0.999903, which raises the statistic from 39.569 to 39.573 - a change in the fourth decimal place that alters no conclusion. Reporting the tie handling still matters, because the size of the adjustment is a fact about this dataset and not a general property.

Solution 13.8

Removing two low scores from the low group raises that group's mean and mean rank and shortens its lower tail, which reduces its standard deviation. Removing one flagged value from the moderate group changes it slightly. The high group is untouched and remains lowest.

The consequence is that the three groups become *more* separated, not less. The verified values confirm it: the low group's mean rises from 56.420 to 57.629 and its standard deviation falls from 9.461 to 7.755; the overall statistic rises from 39.573 to 41.560; jamovi epsilon-squared rises from 0.302 to 0.325; and the bias-adjusted rank effect size rises from 0.291 to 0.314. Deletion made the finding stronger. Anyone who assumed that removing outliers is a cautious act has just been shown otherwise.

Level 3 Solutions

Solution 13.9

$3029.0 + 3999.5 + 1749.5 = 8778.0$. The total of the ranks 1 through 132 is $132 \times 133 / 2 = 8778$. They agree.

The check verifies that every observation received exactly one rank and that tied observations were assigned average ranks summing correctly - in other words, that the ranking step was performed on the intended 132 rows. It does not verify that the groups were assigned correctly, that the right variable was ranked, or that the test statistic was computed properly. It is an arithmetic guard, not a validation of the analysis.

Solution 13.10

Mean rank = $3999.5 / 44 = 90.898$. Under no association, every group's mean rank would be near the average of all ranks, $(N + 1) / 2 = 133 / 2 = 66.5$. The moderate group sits 24.4 rank positions above that reference, and it sits above it while the high group sits 26.7 below it. The two departures are of similar magnitude in opposite directions, which is the numerical signature of the non-monotonic pattern.

Solution 13.11

jamovi $\epsilon^2 = 39.573 / (132 - 1) = 39.573 / 131 = 0.302$.

The bias-adjusted rank effect size is $(39.573 - 3 + 1) / (132 - 3) = 37.573 / 129 = 0.291$. The second formula subtracts $k - 1$ because that is the expected value of H when there is no association at all: with k groups, the statistic follows approximately a chi-square distribution with $k - 1$ degrees of freedom under the null, and a chi-square variable has expectation equal to its degrees of freedom. Subtracting it centres this adjusted measure at zero when nothing is happening. The two quantities must be named separately rather than assigning both formulas the same label.

Solution 13.12

$r = 2 \times 1694.0 / (44 \times 44) - 1 = 3388 / 1936 - 1 = 1.750 - 1 = 0.750$.

In plain language: if you pick one student at random from the moderate-use group and one at random from the high-use group, the moderate student has the higher wellbeing score in about 87.5 percent of such pairs and the

high student in about 12.5 percent, with ties split. The correlation of 0.750 is the difference between those two proportions. It is a statement about pairs, not about points on the wellbeing scale.

Level 4 Solutions

Solution 13.13

The moderate-use group has the highest scores and the tightest distribution: a median of 62.10, an interquartile range of 9.325, and a minimum of 47.9 that is not far below its box. The high-use group is shifted downward as a whole - its third quartile of 55.325 sits below the moderate group's median - and its spread is similar. The low-use group's box overlaps the other two, but it carries a much longer lower tail, reaching 26.9 against a first quartile of 51.8.

The low-use group's mean and standard deviation are the least representative of its typical member. Its standard deviation of 9.461 is 26 to 32 percent larger than either other group's, yet its interquartile range of 9.875 is *smaller* than the high group's 10.550 and only about 6 percent larger than the moderate group's 9.325. A spread statistic that reacts strongly to the tails and a spread statistic that ignores them point in opposite directions; that disagreement is the diagnostic, and it says the standard deviation is being driven by observations outside the central half of the data.

Solution 13.14

Wellbeing scores differed across the three app-use groups, $\chi^2(2, N = 132) = 39.57, p < .001$, jamovi $\epsilon^2 = 0.302$ (bias-adjusted rank effect size = 0.291). Mean ranks were highest in the moderate-use group (90.90), intermediate in the low-use group (68.84), and lowest in the high-use group (39.76), so the pattern across the ordered usage levels rises and then falls rather than running in a single direction.

Study in the Larger Evidence Program: Wellbeing and Reported App Use

- **What this study adds:** A reproducible rank-based estimate of how wellbeing orders across three self-reported usage levels, together with a demonstration that the pattern is not monotonic.
- **What it cannot settle:** Whether app use affects wellbeing, whether wellbeing affects app use, or whether a third factor produces both. Cross-sectional data contain no temporal ordering.
- **Causal-support role:** Weak consistency evidence at best. No responsiveness evidence is present because nothing was deliberately changed, and no mechanism was measured.
- **Next evidence priority:** Objective usage logging with repeated wellbeing measurement, so that within-person change in usage and in wellbeing can be observed in sequence rather than inferred from one occasion.

Solution 13.15

The rank-biserial correlation is negative for low versus moderate (-0.359), so the moderate group sits higher. It is positive for low versus high (0.465) and for moderate versus high (0.750), so the low and moderate groups both sit above the high group.

The moderate and high groups are furthest apart, with a rank-biserial of 0.750 - the largest of the three. The low and moderate groups are closest, with a magnitude of 0.359 .

Holm's adjustment multiplies the smallest p value by the number of comparisons, the next by one fewer, and so on, taking a running maximum. The largest p value in the family is multiplied by one and is therefore unchanged, which is why the low-versus-moderate comparison shows the same value at .004 before and after adjustment. The other two, being smaller, are multiplied by three and two respectively. Holm adjusts every comparison in the family, including the one whose value does not move.

Solution 13.16

H rose from 39.573 to 41.560, jamovi epsilon-squared rose from 0.302 to 0.325, and the bias-adjusted rank effect size rose from 0.291 to 0.314. The low group's mean rose from 56.420 to 57.629 and its standard deviation fell from 9.461 to 7.755. Every quantity moved in the direction of a larger, tidier, more separated effect.

"Removed to be conservative" is untenable because it claims a direction the numbers contradict. Deleting the flagged observations removed the cases that were pulling the low group down toward the high group, which increased the separation between groups and shrank the within-group variability that the test compares it against. Deletion is not conservative or liberal by nature; its direction depends entirely on where the deleted points sat, which is precisely why the rule must be fixed before anyone looks.

Level 5 Solutions

Solution 13.17

A Kruskal-Wallis test on the three independent groups is the defensible choice, followed by prespecified pairwise Mann-Whitney comparisons with a multiplicity adjustment. The justification is the shape of the outcome: a long lower tail in one group and extreme low values that make the arithmetic mean a poor summary of a typical student. The question is about which group's students tend to score higher, and a rank method estimates exactly that.

"The variances were unequal" would be a false justification. The three standard deviations are 9.461, 7.167, and 7.479, and a median-centred Brown-Forsythe test gives $F(2, 129) = 0.54$, $p = .583$ - no evidence of unequal spread. Two things follow. First, the stated reason must match the inspected evidence, or the reader cannot audit the decision. Second, unequal variance would not have been a reason for a rank method anyway: the Kruskal-Wallis test is not a remedy for heteroscedasticity, and when spreads differ substantially it can detect that difference in spread rather than the difference in location the analyst intended to study.

Solution 13.18

The sentence attributes to the test a quantity the test does not target. Kruskal-Wallis compares whether observations in one group tend to sit higher in the combined ordering; medians are not in the statistic. The claim becomes correct only under the additional assumption that the three distributions have the same shape and differ by a location shift - an assumption that must be inspected and stated, and one that the differing tail lengths here make debatable.

Defensible replacement: "Wellbeing scores tended to differ across app-use groups, $\chi^2(2, N = 132) = 39.57$, $p < .001$, jamovi $\epsilon^2 = 0.302$. Group medians were 55.65, 62.10, and 49.00 for the low, moderate, and high groups respectively." The test result and the descriptive medians are both reported, and neither is presented as the other.

Solution 13.19

The verified mean ranks are 68.84 for low, 90.90 for moderate, and 39.76 for high. The pattern rises by 22 rank positions and then falls by 51. A linear trend model fitted across the ordered levels would summarise those opposite movements as one slope, and would describe the data badly no matter which slope it produced. A trend test can also lose power precisely where a non-monotonic pattern is strongest, so the appeal to power fails on its own terms.

The proposal would have been sensible if a monotonic dose-response relationship had been the prespecified hypothesis and the inspected data were consistent with it. Choosing a trend test *after* seeing that the grouping variable is ordinal, without checking whether the pattern is monotonic, substitutes a property of the variable's label for a property of the data.

Solution 13.20

First, the family of comparisons is defined by the analysis plan, not by the results. Omitting one comparison after seeing it makes the reported Holm adjustment wrong, because Holm's procedure depends on the number

of comparisons in the family; two reported values adjusted as if from a family of three are not interpretable as either.

Second, “in the same direction anyway” is a claim about the result, and using it as a reason to omit the comparison means the reporting decision was made after and because of the outcome. A reader cannot then distinguish the reported family from a family selected to look tidy.

What should have been prespecified: which pairwise comparisons would be made, in what direction, with which effect size, and under which multiplicity adjustment - all recorded before the pairwise output was produced.

Study in the Larger Evidence Program: The Cleaned Analysis

- **What this study adds:** A concrete demonstration that deleting flagged observations is not a conservative act. Removing three flagged values raised the statistic from 39.573 to 41.560, jamovi epsilon-squared from 0.302 to 0.325, and the bias-adjusted rank effect size from 0.291 to 0.314.
- **What it cannot settle:** Anything the original analysis could not settle. Deletion changes the sample, never the design.
- **Causal-support role:** None. The reduced analysis is observational for exactly the same reasons as the full one, and its population is now defined partly by the outcome.
- **Next evidence priority:** A prespecified data-handling plan written before collection, naming exclusion rules and the sensitivity analyses that will be reported alongside them.

Level 6 Solutions

Solution 13.21

Wellbeing scores were compared across three self-reported app-use groups ($n = 44$ per group). Distributions were asymmetric, with a long lower tail in the low-use group (minimum 26.9 against a first quartile of 51.8), so a rank-based comparison was prespecified. Scores differed across groups, $\chi^2(2, N = 132) = 39.57, p < .001$, jamovi $\epsilon^2 = 0.302$ (bias-adjusted rank effect size = 0.291), with mean ranks of 68.84, 90.90, and 39.76 for the low, moderate, and high groups. All three pairwise Mann-Whitney comparisons were conducted with Holm adjustment: moderate exceeded low ($U = 620.5, p = .004$, rank-biserial = -0.359 in the low-minus-moderate direction), low exceeded high ($U = 1418.5, p < .001$, rank-biserial = 0.465), and moderate exceeded high ($U = 1694.0, p < .001$, rank-biserial = 0.750). Ties received average ranks and the tie correction factor was 0.9999. Because app-use group was self-reported rather than assigned, these results describe an observed ordering and do not establish that app use affects wellbeing.

Solution 13.22

The arithmetic is right: $61.834 - 49.702 = 12.13$. Two things are nevertheless wrong.

The first is a scale mismatch. The 12.13-point figure is a difference between arithmetic means, and it is being reported as though the Kruskal-Wallis statistic tested it. It did not. The rank-based analysis produced no quantity in wellbeing points, and the mean difference it is attached to was never subjected to the test whose statistic follows it.

The second is that attaching a group-level descriptive difference to an omnibus test across three groups implies that this specific pair was tested by that statistic. It was not; the pairwise comparison is a separate analysis with its own result.

Honest version: “Wellbeing scores tended to differ across app-use groups, $\chi^2(2, N = 132) = 39.57, p < .001$, jamovi $\epsilon^2 = 0.302$. The moderate-use and high-use groups differed in the prespecified pairwise comparison, $U = 1694.0$, Holm-adjusted $p < .001$, rank-biserial = 0.750; descriptively, their means were 61.83 and 49.70 and their medians 62.10 and 49.00.”

Solution 13.23

To the methods: “Three observations carrying the file’s outlier flag were removed before analysis: two from the low-use group and one from the moderate-use group. The removal rule was applied after the data were inspected and was not prespecified.”

To the results: “The analysis of the complete file gave $\chi^2(2, N = 132) = 39.57, p < .001$, jamovi $\epsilon^2 = 0.302$ (bias-adjusted rank effect size = 0.291); the analysis after removal gave $\chi^2(2, N = 129) = 41.56, p < .001$, jamovi $\epsilon^2 = 0.325$ (bias-adjusted rank effect size = 0.314).”

The complete-file analysis should be the primary result. It estimates the quantity that was defined before anyone looked at the data, on the sample that was actually collected. The reduced analysis is a sensitivity check reported alongside it. Reversing that order would make an unplanned, outcome-dependent exclusion the basis of the headline claim, and would leave the reader unable to see that the exclusion increased the effect.

Solution 13.24

“App-use group was self-reported at a single occasion and was not assigned, so the three groups may differ in many ways besides how much they use the app. The design contains no temporal ordering, so it cannot distinguish an effect of app use on wellbeing from an effect of wellbeing on app use, nor from a third factor that influences both. The rank-based statistics describe how the groups order relative to one another and how strongly group membership accounts for that ordering; they do not estimate a difference in wellbeing points, and the group means and medians reported alongside them are descriptive summaries rather than tested quantities.”

PART V

Designs that do not randomize

Unit 14: Quasi-Experimental Comparison and Covariate Adjustment

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Unit Purpose

Some groups cannot be assigned, even in principle. A child cannot be randomized into an age band, and no improvement in funding, ethics review, or sample size will change that. A quasi-experimental comparison is therefore not an experiment with a missing feature; it is a different kind of study whose claim ceiling is set by what the grouping variable is. Adding a covariate does not raise that ceiling. It changes which comparison the coefficient describes, and whether that change helps depends on where the covariate sits in the causal order. This unit develops the habit of asking what a covariate *is* before asking what adjusting for it does.

Learning Objectives

After completing this unit, a learner should be able to:

- distinguish a grouping variable that was assigned from one that was observed, and one that cannot be assigned at all;
- read an analysis of variance table and identify the quantity each row contributes;
- distinguish a multiplicity adjustment from a covariate adjustment, both of which are called “adjustment”;
- explain what a covariate-adjusted group coefficient estimates and how it differs from the unadjusted difference;
- decide whether adjusting for a variable reduces bias or removes part of the effect of interest, using the causal order rather than the fit statistics;
- check the conditions an analysis of covariance requires, including comparable slopes across groups; and
- write a limitation that names the specific alternative explanations a quasi-experimental design leaves open.

Decision Map

Ask these questions in order:

1. What is the scientific question, and is the grouping variable assigned, observed, or unassignable?
2. What are the cases, and how many observations does each contribute?
3. What is the unadjusted comparison, and what does it estimate?
4. For each candidate covariate: was it measured before the grouping variable, caused by it, or neither?
5. What quantity would the adjusted coefficient estimate, and is that the quantity the question asked for?
6. What conditions does the adjusted model require, and does the evidence support them?
7. What does the design identify, what is the claim ceiling, and what evidence should come next?

Study Files

The primary dataset is `study_10_developmental_emotion_recognition.csv`. The teaching variant is a subset of the same file, reproducible in jamovi with a row filter on `age_group`. Public CSV files, dictionaries, and verified result records are available at <https://github.com/nicholaskarlson/data>.

Study	Data structure	Primary outcome	Unit role
Study 10	144 children, 48 in each of three age bands	Emotion recognition accuracy	Three-group quasi-experimental comparison with a covariate
Study 10A	The 96 children in the 6-7 and 10-11 bands	Emotion recognition accuracy	Two-group contrast; the covariate imbalance made visible

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each Pause and Decide, stop reading and write an answer, reason, and confidence rating.

Plausible Research Scenario: Emotion Recognition Across Three Age Bands

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A fictional developmental research group tests 144 children on a task that measures how accurately they identify emotional expressions. Forty-eight children are in each of three age bands: 6 to 7, 8 to 9, and 10 to 11 years. Each child completes the task once and contributes one accuracy score. A vocabulary score is also recorded for every child.

The research question is: **how does emotion recognition accuracy differ across these three age bands?**

Age band was not assigned and could not have been. The children arrived at their ages by living, and every difference that accompanies being older - schooling, language, social experience, attention span, familiarity with tests - travels with the grouping variable.

Pause and Decide

Before reading further, name the case, state whether the design is experimental, decide what quantity the three-group comparison estimates, and write down what the phrase “the effect of age” could and could not mean here. Then decide, before you see any output, whether you would adjust for vocabulary and say why.

Why This Model?

A one-way analysis of variance across the three age bands estimates the differences among the three group means, and prespecified pairwise comparisons with a multiplicity adjustment locate where those differences sit. The model matches the data structure: one score per child, three independent groups, a quantitative outcome.

The model is correctly specified and the design identifies nothing causal. Those two statements are compatible and are the heart of this unit. The analysis estimates how accuracy differs across age bands in the represented sample. It does not isolate a mechanism, because no mechanism was manipulated and none could be.

Multiplicity matters because three pairwise comparisons are a family. The unit uses Bonferroni adjustment, which multiplies each comparison’s p value by the number of comparisons and widens the confidence intervals to match. The word “adjustment” here means something entirely different from what it will mean in the second scenario, and the difference is worth fixing in mind now: a multiplicity adjustment changes how a family of results is judged, and changes no estimate; a covariate adjustment changes the estimate itself.

What Would a Favorable Result Look Like?

The group would hope to see accuracy increasing across the age bands with intervals precise enough to distinguish adjacent bands from one another. A result in which the two older bands cannot be distinguished would also be informative: it would suggest that whatever changes between 6 and 9 has largely finished by 9, which is a substantive finding about the shape of development rather than a failure.

Best Defensible Claim

Statistical evidence: The analysis estimates differences in mean accuracy across three age bands, with intervals and a multiplicity-adjusted family of pairwise comparisons.

Causal identification: None beyond the association. Age band cannot be assigned, so nothing separates chronological age from everything that accompanies it.

Claim ceiling: The group may describe how accuracy differs across age bands among the represented children. It may not claim that age *causes* the difference in any sense that isolates age from schooling, language, or experience, and it may not describe the pattern as within-child development, because no child was measured twice.

What Must Be True?

Each child contributes one independent score. The task must measure the same construct comparably at all three ages - a stronger requirement than it looks, because an instrument can be harder to understand for younger children in ways unrelated to emotion recognition. For the pooled-variance model, spreads should be reasonably similar and residuals reasonably well behaved.

Defensibility Procedures

Prespecify the outcome, the comparisons, and the multiplicity adjustment. Report all three pairwise comparisons with adjusted *p* values and family-wise intervals, not only the ones that cross a threshold. Report the assumption checks. State explicitly that the design is cross-sectional and that age band is not an assigned condition.

Next evidence priority: A longitudinal design measuring the same children more than once, which would replace between-child differences with within-child change and would separate age from cohort.

Claim Audit

Repair this claim before continuing: “Being older causes a 13-point improvement in emotion recognition.” Your repair should address both the causal language and the phrase “improvement,” which implies within-child change that this design never observed.

Plausible Research Scenario: The Analyst Who Controlled for Vocabulary

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A second analyst observes that vocabulary scores differ across the three age bands and that vocabulary correlates with accuracy. Concerned that the age comparison is “confounded by language ability,” the analyst adds vocabulary as a covariate and reports the adjusted age coefficients as the corrected result.

The adjusted coefficients are smaller than the unadjusted differences. The analyst reports this as evidence that part of the apparent age effect was really a vocabulary effect.

Study 10A restricts the same file to the youngest and oldest bands so that the two quantities can be compared directly.

Pause and Decide

Before reading further, decide what vocabulary *is* in the causal order of this study. Was it measured before age band? Could age band have caused it? Could it have caused age band? Then predict what will happen to the age coefficients after adjustment, and decide in advance whether a reduction would be good news or bad news.

Why This Model?

Analysis of covariance is a correctly specified regression model here. It fits, its residuals behave, and its coefficients have small standard errors. None of that tells the analyst whether the coefficients answer the question.

The decisive fact is the causal order. Age can cause vocabulary; vocabulary cannot cause age. Vocabulary is therefore a **consequence of the grouping variable**, not a pre-existing confounder. Adjusting for a consequence removes part of the very effect being estimated. The adjusted age coefficient answers a different question: “how much do accuracy scores differ between age bands among children who happen to have the same vocabulary score?” That is a legitimate question - it is roughly a question about the part of the age difference that does not travel through vocabulary - but it is not the total developmental difference, and it must not be labelled a corrected version of it.

The general rule is worth stating plainly. Adjustment reduces bias when the covariate is a common cause of the grouping variable and the outcome, measured before both. Adjustment introduces bias, or answers a different question, when the covariate is caused by the grouping variable, or is caused by both the grouping variable and the outcome. No fit statistic distinguishes these cases. Only an argument about the causal order does, and that argument comes from subject knowledge, not from the data.

What Would a Favorable Result Look Like?

There is no favorable result here, only a correctly labelled one. Reporting both the unadjusted difference and the adjusted coefficient, saying what each estimates, is the outcome to aim for.

Best Defensible Claim

Statistical evidence: The unadjusted comparison estimates the total difference in accuracy between age bands. The adjusted coefficient estimates the difference remaining after holding vocabulary fixed, which is a different quantity.

Causal identification: Adjustment does not create identification. The design is quasi-experimental before and after.

Claim ceiling: The analyst may report both quantities and name each. The analyst may not present the adjusted coefficient as a corrected or purified age effect, and may not claim to have removed confounding by adjusting for a variable that age produces.

What Must Be True?

For the model to be interpretable at all, the relationship between vocabulary and accuracy should be similar in each age band, so that a single covariate slope is a reasonable summary. That condition can be tested by adding an interaction between age band and vocabulary. Passing the test does not make the adjustment appropriate; it only removes one reason it would be uninterpretable.

Defensibility Procedures

State the causal role assumed for every covariate before fitting. Report unadjusted and adjusted estimates together. Report the covariate’s own coefficient and the group differences on the covariate, so the reader can see how much the groups differ on it. Test and report the homogeneity of slopes. Never describe an adjusted estimate as the “true” effect.

Next evidence priority: A design in which a genuinely manipulable variable is assigned - for example, a task instruction or a training exposure - so that a causal contrast exists to estimate.

Claim Audit

Explain why “after controlling for vocabulary, age still has a significant effect” is a misleading sentence in this study even though every number in the underlying model is correct.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 14.1 — Classify the grouping variable

Age band in Study 10 is not randomly assigned. Distinguish three cases: a variable that was not assigned but could have been, a variable that was not assigned and could not have been, and a variable that was assigned. Place age band, message condition from Unit 12, and app-use group from Unit 13 into those categories, and say which category has the lowest claim ceiling.

Problem 14.2 — Name the two adjustments

This unit uses the word “adjustment” for two different operations. Name both, state what each one changes, and state what each one leaves unchanged.

Problem 14.3 — State what an ANCOVA coefficient estimates

Write, in one sentence and without formulas, what the coefficient on `age_10_11` estimates in a model that also contains vocabulary. Then state what it does *not* estimate.

Problem 14.4 — Recognize the covariate’s causal position

Vocabulary was recorded at the same session as the accuracy task. State whether it could be a common cause of age band and accuracy, whether it could be a consequence of age band, and what follows for the interpretation of an adjusted age coefficient.

Level 2: Predict Before Clicking

Problem 14.5 — Predict the direction and rough size

The three group means for accuracy are 59.354, 69.300, and 72.360 for the 6-7, 8-9, and 10-11 bands. Predict which pairwise contrast will be largest, which will be smallest, and whether the smallest one will survive a Bonferroni adjustment across three comparisons.

Problem 14.6 — Predict the covariate imbalance

The three group means for vocabulary are 41.269, 47.967, and 53.710. Predict whether vocabulary is balanced across age bands, and predict what that imbalance implies for how much the age coefficients will move when vocabulary enters the model.

Problem 14.7 — Predict the direction of the adjustment

Vocabulary is higher in older bands and is positively related to accuracy. Predict whether adjusting for vocabulary will make the age coefficients larger or smaller than the unadjusted differences, and explain the mechanism in one sentence.

Problem 14.8 — Predict what adjustment cannot do

Before seeing any output, state one thing the adjusted model will improve and one thing it cannot improve. Be specific: name a statistic for the first and a design property for the second.

Level 3: Calculate and Explain

Problem 14.9 — Complete the ANOVA table

The between-groups sum of squares is 4439.173 with 2 degrees of freedom; the within-groups sum of squares is 7274.354 with 141 degrees of freedom. Compute both mean squares and the F ratio, and state what the denominator mean square represents.

Problem 14.10 — Compute the effect size and compare two versions

Using SS-between 4439.173 and SS-total 11713.527, compute eta-squared. The verified omega-squared is 0.369. Explain in one sentence why omega-squared is smaller and which of the two is the less optimistic estimate of the population value.

Problem 14.11 — Recover a pairwise standard error and statistic

The pooled within-groups mean square is 51.591 and each group contains 48 children. Compute the standard error of a difference between two group means using $SE = \sqrt{MS\text{-within} \times (1/n_1 + 1/n_2)}$. Then compute the t statistic for the 8-9 versus 10-11 contrast, whose mean difference is -3.060 .

Problem 14.12 — Compare an adjusted and an unadjusted difference

The unadjusted difference between the 6-7 and 10-11 bands is 13.006 points. The coefficient on age_10_11 in the model containing vocabulary is 8.917. Compute the reduction in points and as a percentage of the unadjusted difference, and state which of the two numbers answers the question “how much does accuracy differ between six-year-olds and ten-year-olds?”

Level 4: Read jamovi Output**Problem 14.13 — Read the ANOVA and its checks**

Source	SS	df	MS	F	p	η^2	ω^2
Age band	4439.173	2	2219.586	43.02	< .001	0.379	0.369
Residual	7274.354	141	51.591				

A median-centred Brown-Forsythe test gives $F(2, 141) = 0.77$, $p = .467$, and a Shapiro-Wilk test on the residuals gives $W = 0.988$, $p = .247$. Write one sentence reporting the omnibus result and one sentence stating what the two checks do and do not license.

Problem 14.14 — Read the pairwise family

Contrast	Mean difference	SE	t	Raw p	Bonferroni p	95% family-wise CI
6-7 vs 8-9	-9.946	1.466	-6.78	< .001	< .001	[-13.498, -6.393]
6-7 vs 10-11	-13.006	1.466	-8.87	< .001	< .001	[-16.559, -9.454]
8-9 vs 10-11	-3.060	1.466	-2.09	.039	.116	[-6.613, 0.492]

Describe what happened to the third contrast and explain why its interval contains zero while its raw p value does not exceed .05. State what a reader should conclude about the two older bands.

Problem 14.15 — Read the covariate model

Term	b	SE	t	p	95% CI	β
Intercept	45.791	3.500	13.08	< .001	[38.871, 52.712]	
Age 8-9	7.745	1.495	5.18	< .001	[4.788, 10.701]	0.405
Age 10-11	8.917	1.722	5.18	< .001	[5.513, 12.321]	0.466
Vocabulary	0.329	0.081	4.04	< .001	[0.168, 0.490]	0.315

The model has $R^2 = 0.444$, adjusted $R^2 = 0.432$, $F(3, 140) = 37.23$, $p < .001$. The reference band is 6-7. Read the vocabulary coefficient in its own units, then state what the age_10_11 coefficient of 8.917 describes and how it differs from the 13.006 reported in the previous table.

Problem 14.16 — Read the slope-homogeneity check

Adding an age-band by vocabulary interaction to the model gives $F(2, 138) = 0.67, p = .516$. The within-band correlations between vocabulary and accuracy are 0.221, 0.403, and 0.322 for the 6-7, 8-9, and 10-11 bands. State what the interaction test licenses, and explain why the three differing correlations are not in conflict with it.

Level 5: Choose the Model

Problem 14.17 — Choose the primary analysis

The question is how accuracy differs across three age bands, with one score per child. Choose the primary analysis and the follow-up procedure, and justify both from the data structure. Then state what you would report as the primary estimate and why.

Problem 14.18 — Reject the covariate as a correction

An analyst argues that the adjusted coefficient of 8.917 is the “real” age effect because the unadjusted 13.006 is contaminated by vocabulary. Reject the argument. Your rejection must identify the causal position of vocabulary and say what quantity 8.917 does estimate.

Problem 14.19 — Reject the reversed adjustment

A different analyst proposes running the model the other way around: predicting vocabulary from age band while adjusting for emotion accuracy, “to see which one is really driving it.” Explain why this cannot arbitrate the question, and name the general error the proposal commits.

Problem 14.20 — Choose what to do about the third contrast

The 8-9 versus 10-11 contrast has a raw p of .039 and a Bonferroni-adjusted p of .116, with a family-wise interval of $[-6.613, 0.492]$. An analyst proposes reporting the raw p value because the comparison “was of primary interest all along.” Decide whether that is defensible, state the condition under which it would be, and write the sentence that should be reported either way.

Level 6: Report and Critique

Problem 14.21 — Write the result paragraph

Using the verified values, write an APA-style results paragraph for the unadjusted analysis of Study 10. Include group descriptives, the omnibus test with effect size, the multiplicity-adjusted pairwise family, the assumption checks, and one sentence on what the design permits.

Problem 14.22 — Write the paragraph that reports both models honestly

Write the short paragraph that reports the unadjusted and adjusted analyses together. It must name what each quantity estimates, must not describe either as the corrected version of the other, and must include the covariate imbalance across bands.

Problem 14.23 — Repair a within-child claim

Repair this sentence: “Emotion recognition improved by 13.01 points between ages six and ten, $F(2, 141) = 43.02, p < .001$.” Identify every distinct error, including one about what was measured and one about which comparison the statistic refers to.

Problem 14.24 — Write the quasi-experimental limitation

Write the three-sentence limitation for Study 10. One sentence must name at least two specific alternative explanations that travel with age band; one must address the cross-sectional structure; one must address the measurement instrument.

Solutions

Level 1 Solutions

Solution 14.1

Message condition in Unit 12 was assigned. App-use group in Unit 13 was not assigned but could in principle be assigned in a different study, since a researcher could allocate people to a usage regimen. Age band in Study 10 was not assigned and cannot be: no procedure allocates a child to being seven.

The lowest claim ceiling belongs to the unassignable variable, and it is lowest in a permanent way. An observational grouping like app use can be raised toward a causal claim by a future randomized study; age band cannot, because there is no counterfactual in which the same child is simultaneously six and ten at one moment. Studies of age therefore change what they estimate - moving to within-child change over time - rather than improving how they estimate the same thing.

Solution 14.2

Multiplicity adjustment changes the criterion applied to a family of comparisons. It rescales p values and widens intervals to control the family-wise error rate. It leaves every point estimate unchanged: the mean difference of -3.060 is the same number before and after Bonferroni.

Covariate adjustment changes the estimate itself. Adding vocabulary to the model changes the age coefficient from 13.006 to 8.917 because it is now estimating a different quantity. It says nothing about error rates.

Sharing one English word for these two operations is a persistent source of confusion, and a results section should make clear which is meant.

Solution 14.3

The coefficient on `age_10_11` estimates the average difference in accuracy between children in the 10-11 band and children in the 6-7 reference band, among children with the same vocabulary score.

It does not estimate the total difference in accuracy between children of those two ages, because part of that total difference operates through the vocabulary that older children have.

Solution 14.4

Vocabulary cannot be a common cause of age band and accuracy, because nothing can cause a child's age. It can be, and in a developmental study almost certainly is, a consequence of age band: older children have larger vocabularies partly because they are older.

What follows is that adjusting for vocabulary conditions on a variable that lies between the grouping variable and the outcome. The adjusted coefficient therefore estimates the portion of the age difference that does not operate through vocabulary, rather than a de-confounded age effect. Calling it "controlling for" a confounder misdescribes both the operation and the result.

Level 2 Solutions

Solution 14.5

The largest contrast is 6-7 versus 10-11, at $72.360 - 59.354 = 13.006$ points. The smallest is 8-9 versus 10-11, at $72.360 - 69.300 = 3.060$ points. With a standard error near 1.47 , the smallest contrast gives a t near 2.1 , which is around the two-sided $.05$ boundary unadjusted and will not survive multiplication by three. The verified values confirm it: raw $p = .039$, Bonferroni $p = .116$.

Solution 14.6

Vocabulary is strongly unbalanced: 41.269 , 47.967 , and 53.710 , a spread of 12.44 points across bands, and a one-way analysis of variance on vocabulary alone gives $F(2, 141) = 37.36$, $p < .001$, $\eta^2 = 0.346$.

A covariate that differs this much across groups and is related to the outcome will move the group coefficients substantially when it is added. That is a statement about arithmetic, not about correctness: large movement is what happens when the covariate carries much of the same information as the grouping variable, and it is equally consistent with removing confounding and with removing part of the effect.

Solution 14.7

The coefficients will get smaller. Older bands have higher vocabulary, and higher vocabulary is associated with higher accuracy, so part of the raw accuracy gap between bands is accompanied by the vocabulary gap. Holding vocabulary fixed removes that accompanying part, leaving a smaller age coefficient. The verified reduction is from 13.006 to 8.917 for the oldest band.

Solution 14.8

The adjusted model will improve the residual mean square, and with it the precision of the estimates: the residual mean square falls from 51.591 to 46.540 when vocabulary is added, because vocabulary explains variation that was previously unexplained.

It cannot improve identification. Age band remains unassigned and unassignable after adjustment exactly as before, so no coefficient in the adjusted model is a causal effect of age. A better-fitting model on an unidentified design is a more precise estimate of an associational quantity.

Level 3 Solutions

Solution 14.9

$MS\text{-between} = 4439.173 / 2 = 2219.586$. $MS\text{-within} = 7274.354 / 141 = 51.591$. $F = 2219.586 / 51.591 = 43.02$.

The denominator mean square represents variation among children *within* age bands - differences between children of the same age. It is the yardstick against which between-band variation is judged: the F ratio asks how large the differences among band means are relative to the differences among children who share a band.

Solution 14.10

$\eta^2 = 4439.173 / 11713.527 = 0.379$.

Omega-squared is smaller (0.369) because it subtracts the variation the between-groups sum of squares would be expected to contain even if the population means were identical. Eta-squared describes the sample and is biased upward as an estimate of the population value; omega-squared corrects for that bias and is the less optimistic of the two. With 48 children per band the correction is small; in a small study the gap can be substantial.

Solution 14.11

$SE = \sqrt{(51.591 \times (1/48 + 1/48))} = \sqrt{(51.591 \times 0.0416667)} = \sqrt{2.14963} = 1.466$.

$t = -3.060 / 1.466 = -2.087$, on 141 degrees of freedom. Note that the standard error uses the pooled within-groups mean square from all three bands, not only the two being compared, which is why every pairwise contrast in this balanced design shares the same standard error of 1.466.

Solution 14.12

Reduction = $13.006 - 8.917 = 4.089$ points, which is $4.089 / 13.006 = 31.4$ percent of the unadjusted difference.

The question “how much does accuracy differ between six-year-olds and ten-year-olds?” is answered by 13.006. That is the total observed difference between the two bands. The 8.917 answers a narrower question about children matched on vocabulary, which is not a group of children that the original question was about. Neither number is wrong; using one to answer the other’s question is.

A useful cross-check: restricting the file to those two bands alone (Study 10A) reproduces the same 13.006-point difference, with Welch's $t(91.84) = 9.29$, $p < .001$, 95 percent CI [10.23, 15.79], Hedges' $g = 1.88$. The point estimate is identical because it is the same two group means; only the standard error differs, because the three-group analysis pools within-band variability from all 144 children (SE 1.466) while the two-group analysis uses only the 96 in those bands (SE 1.400), and because the interval reported in the pairwise family is widened for multiplicity while this one is not. Study 10A also makes the covariate imbalance vivid: across those same two bands, vocabulary differs by 12.44 points.

Level 4 Solutions

Solution 14.13

Emotion recognition accuracy differed across the three age bands, $F(2, 141) = 43.02$, $p < .001$, $\eta^2 = 0.379$, $\omega^2 = 0.369$.

The two checks license the use of the pooled-variance model and the F reference distribution: spreads are similar across bands ($F = 0.77$, $p = .467$) and the residuals show no departure from normality worth acting on ($W = 0.988$, $p = .247$). They license nothing about the design. Neither check says anything about whether age band was assigned, whether the task measures the same construct at all three ages, or whether the sample represents any wider population. Assumption checks concern the model; they are silent about identification.

Study in the Larger Evidence Program: Age and Emotion Recognition

- **What this study adds:** A precise cross-sectional description of how accuracy differs across three age bands in one sample, with a covariate whose imbalance is documented.
- **What it cannot settle:** Whether the difference reflects maturation, schooling, language, test familiarity, or cohort, and whether any individual child improves - no child was measured twice.
- **Causal-support role:** Consistency evidence only. No responsiveness evidence is possible, because the grouping variable cannot be manipulated.
- **Next evidence priority:** A longitudinal design following the same children across the same age range, which converts between-child differences into within-child change and removes the cohort explanation.

Solution 14.14

The third contrast lost its threshold status under multiplicity adjustment: raw $p = .039$ became Bonferroni $p = .116$, and the family-wise 95 percent interval [-6.613, 0.492] includes zero.

There is no contradiction. The interval reported here is a *family-wise* interval, widened by the same adjustment that multiplied the p value, so that all three intervals hold simultaneously at 95 percent confidence. An unadjusted 95 percent interval for the same contrast would have been narrower and would have excluded zero, matching the raw p value. The interval and the p value agree with one another as long as both are adjusted or both are not.

A reader should conclude that the two older bands could not be reliably distinguished within a family of three comparisons, and that the data are compatible with a difference between them of up to about 6.6 points in one direction or about half a point in the other. That is a statement about precision, not a demonstration that the two bands are equivalent.

Solution 14.15

The vocabulary coefficient of 0.329 means that among children in the same age band, each additional vocabulary point is associated with about a third of a point more accuracy, with a 95 percent interval of [0.168, 0.490].

The age_10_11 coefficient of 8.917 describes the difference in accuracy between the 10-11 band and the 6-7 reference band *among children with the same vocabulary score*. The 13.006 in the previous table describes the

difference between those same two bands with no such restriction - the total difference between the groups as they actually are.

They differ by 4.089 points because older children have higher vocabulary, so holding vocabulary constant removes part of what distinguishes the bands. The adjusted figure is not a corrected version of the unadjusted one. It is an answer to a different question, and in this study the unadjusted figure is the one that matches the research question as stated.

Solution 14.16

The interaction test licenses the use of a single common slope for vocabulary across the three bands: there is no evidence that the vocabulary-accuracy relationship differs by band, $F(2, 138) = 0.67, p = .516$. That removes one reason the adjusted coefficients would be uninterpretable.

The three within-band correlations of 0.221, 0.403, and 0.322 are not in conflict with that result. A correlation depends on both the slope and the spreads of the two variables within the band, so correlations can differ while slopes do not. More importantly, three correlations estimated on 48 children each carry substantial sampling error; differences of this size are entirely ordinary under a common underlying slope, which is exactly what the interaction test reports.

Passing this check does not make the adjustment appropriate. It says the model is internally coherent, not that the quantity it estimates is the quantity anyone wanted.

Level 5 Solutions

Solution 14.17

A one-way analysis of variance across the three age bands, followed by all three pairwise comparisons with a prespecified Bonferroni adjustment, is the defensible primary analysis. The structure supports it: one independent score per child, three groups, a quantitative outcome, similar spreads, and well-behaved residuals.

The primary estimates to report are the three mean differences with their family-wise intervals, because they are on the scale of the outcome and answer the question directly. The omnibus F is worth reporting as the family's gate, but a reader learns more from -9.946 , -13.006 , and -3.060 with intervals than from a single F of 43.02. The unadjusted differences are the primary estimates because the research question asks how accuracy differs across age bands as they actually are.

Solution 14.18

The argument fails on the causal position of the covariate. Vocabulary cannot be a confounder of the age-accuracy relationship, because a confounder must be a common cause of both, and nothing causes a child's age. Vocabulary is a consequence of age. Conditioning on a consequence of the grouping variable removes the portion of the effect that travels through it, so 8.917 is a smaller number not because contamination was removed but because part of the effect was.

What 8.917 does estimate is the difference in accuracy between the 6-7 and 10-11 bands among children with the same vocabulary score - loosely, the part of the age difference that does not operate through vocabulary. That is a defensible quantity to report and can be interesting, particularly to someone asking whether age brings anything beyond language development. It is not the total age difference, and it is not a corrected estimate of it.

The general lesson: "adding a covariate reduced the coefficient, so the covariate was a confounder" is a non-sequitur. Adjusting for a mediator also reduces the coefficient. Adjusting for a collider can increase it, or reverse its sign. The output looks the same in all three cases.

Solution 14.19

Reversing the model cannot arbitrate anything, because both models are descriptions of the same joint distribution and neither contains information about direction. Regression is symmetric in that sense: a well-

fitting model of accuracy given vocabulary and a well-fitting model of vocabulary given accuracy can both exist, and neither one licenses a claim about which variable acts on which.

In this study the proposal is additionally impossible on its face, because age band cannot be an outcome of anything measured at the same session.

The general error is treating a fitted coefficient as evidence about causal direction. Direction comes from design - what was manipulated, what was measured first, what could not have been caused by what - and from subject-matter argument. Fit statistics can tell you a model describes the data; they cannot tell you the arrow points left rather than right.

Solution 14.20

It is defensible only if that contrast was designated the primary comparison *before* the data were examined, and it is then reported as the single primary test with the other two labelled as secondary. Under that plan there is no family of three to adjust for, and the raw p value is the correct one.

It is not defensible here. The three comparisons were planned as a family, and selecting one of them as “of primary interest all along” after seeing which one lost its threshold status is a decision made in response to the result. The reported p value would then describe a procedure different from the one actually used.

The sentence to report either way is the same, and it does not depend on the threshold: “The 8-9 and 10-11 bands differed by 3.06 points, 95 percent family-wise CI $[-6.61, 0.49]$, Bonferroni-adjusted $p = .116$; the data do not distinguish these two bands within a family of three comparisons, and are compatible with differences ranging from a 6.6-point advantage for the older band to a small advantage for the younger.”

Study in the Larger Evidence Program: Adjusting for a Consequence

- **What this study adds:** A clean example of a covariate that is caused by the grouping variable rather than confounding it, with both quantities reported: a total difference of 13.006 points and a vocabulary-held-fixed difference of 8.917.
- **What it cannot settle:** Which of the two is the “real” effect of age. That question is malformed; the two estimate different things, and the design identifies neither causally.
- **Causal-support role:** None added by adjustment. A better-fitting model on an unidentified design is a more precise estimate of an association.
- **Next evidence priority:** A design that assigns something manipulable - a task instruction, a training exposure - so that a causal contrast exists to estimate at all.

Level 6 Solutions

Solution 14.21

Emotion recognition accuracy was compared across three age bands with 48 children in each (6-7: $M = 59.35$, $SD = 6.31$; 8-9: $M = 69.30$, $SD = 7.80$; 10-11: $M = 72.36$, $SD = 7.36$). Accuracy differed across bands, $F(2, 141) = 43.02$, $p < .001$, $\eta^2 = 0.379$, $\omega^2 = 0.369$. In a prespecified family of three pairwise comparisons with Bonferroni adjustment, the 6-7 band scored lower than the 8-9 band by 9.95 points, 95 percent family-wise CI $[-13.50, -6.39]$, adjusted $p < .001$, and lower than the 10-11 band by 13.01 points, CI $[-16.56, -9.45]$, adjusted $p < .001$; the 8-9 and 10-11 bands did not differ reliably, difference 3.06 points, CI $[-6.61, 0.49]$, adjusted $p = .116$. Variances were comparable across bands, Brown-Forsythe $F(2, 141) = 0.77$, $p = .467$, and residuals showed no material departure from normality, $W = 0.988$, $p = .247$. Because age band was neither assigned nor assignable, these results describe cross-sectional differences among the represented children and do not isolate an effect of age.

Solution 14.22

“The 6-7 band scored 13.01 points lower than the 10-11 band, 95 percent family-wise CI $[-16.56, -9.45]$. Vocabulary scores also differed substantially across bands ($M = 41.27$, 47.97, and 53.71 for the three bands,

$F(2, 141) = 37.36, p < .001$), and vocabulary was positively related to accuracy within bands, $b = 0.33$, 95 percent CI [0.17, 0.49]. In a model containing both, the coefficient for the 10-11 band was 8.92, 95 percent CI [5.51, 12.32], which estimates the difference between the two bands among children with the same vocabulary score rather than the total difference between the bands. Because vocabulary develops with age, it is a consequence of the grouping variable rather than a confounder of it, so the second estimate is reported as a decomposition of the first and not as a correction to it.”

Solution 14.23

Four distinct errors.

Within-child change. “Improved” describes a child getting better over time. No child in this study was measured twice; the design compares different children of different ages at one occasion. The word must go.

Causal attribution. “Between ages six and ten” attributes the difference to age, when age band carries schooling, language, test familiarity, and cohort along with it.

Wrong statistic for the stated quantity. The 13.01 figure is one pairwise contrast; $F(2, 141) = 43.02$ is the omnibus test across all three bands. Attaching the omnibus statistic to a specific contrast implies that this contrast was what the statistic tested.

Missing uncertainty and multiplicity. No interval is given, and no indication that the contrast belongs to a family of three.

Repaired: “Children in the 10-11 band scored 13.01 points higher on the task than children in the 6-7 band, 95 percent family-wise CI [9.45, 16.56], Bonferroni-adjusted $p < .001$ within a family of three comparisons. Because different children were tested at each age, this is a between-group difference rather than observed change within children.”

Solution 14.24

“Age band was not assigned and could not be, so the comparison carries with it everything that accompanies being older in this sample - among them years of schooling, language development, familiarity with structured testing, and attention span - and the analysis cannot separate chronological age from any of them. The design is cross-sectional, comparing different children at one occasion, so it describes differences between age groups rather than change within children, and it cannot distinguish an age effect from a cohort difference. The task was administered in the same form at all three ages, and if it is harder to understand for the youngest children for reasons unrelated to emotion recognition, part of the observed difference would reflect the instrument rather than the construct.”

Unit 15: Single-Case Designs and Nonoverlap

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Unit Purpose

A single-case design does not have a small sample. It has a different unit. Forty-two daily observations of one person are forty-two measurements of one case, and the comparison runs within a series rather than between people. The evidence comes from the pattern of level, trend, variability, and overlap across phases, and its credibility comes from replication across cases. This unit develops the habit of reading a series before summarising it, computing a nonoverlap statistic by hand so that its meaning is unmistakable, and refusing to convert repeated observations of one person into a between-groups test.

Learning Objectives

After completing this unit, a learner should be able to:

- identify the case in a single-case design and explain why the number of observations is not the number of independent units;
- read a phase series for level, trend, variability, and immediacy of change;
- explain why a declining baseline strengthens some claims and weakens others;
- compute nonoverlap of all pairs by hand, including the half-credit tie rule, and state what it does and does not measure;
- describe an ordinal outcome with frequencies and medians, and mark any mean as supplemental;
- reject a between-groups test applied to repeated observations of one case, and name the assumption it violates; and
- state the generalization boundary that one case imposes no matter how clean the series looks.

Decision Map

Ask these questions in order:

1. What is the scientific question, and whose behaviour is it about?
2. What is the case, and how many cases are there?
3. What are the phases, how long is each, and in what order did they occur?
4. What does each series look like in level, trend, variability, and immediacy at the phase change?
5. What is the measurement level of each outcome, and which summaries are permitted by it?
6. What nonoverlap or level-change summary answers the question, and what does it leave out?
7. What does the design identify, what is the claim ceiling, and what evidence should come next?

Study Files

The primary dataset is `study_11_single_case_habit_tracking.csv`. The teaching variant is not a different file; it is the same data analysed with a model that does not fit, so that its output can be examined and rejected.

Study	Data structure	Primary outcomes	Unit role
Study 11	One case, 42 consecutive days in three phases of 14	Daily habit count; daily stress rating	Complete single-case series with phase structure
Study 11A	The same 28 days from the baseline and intervention phases	Daily habit count; daily stress rating	Days treated as if they were independent people; the model to reject

The phases run in order: baseline on days 1 to 14, intervention on days 15 to 28, and maintenance on days 29 to 42. `habit_count` is a daily count. `stress_rating` is an ordinal daily self-rating with observed values from 1 to 8, where higher values indicate more stress.

Plausible Research Scenarios

The scenarios create decisions rather than documenting real research. At each Pause and Decide, stop reading and write an answer, reason, and confidence rating.

Plausible Research Scenario: One Person, Forty-Two Days

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

A fictional student works with a counsellor on a daily routine. For two weeks the student records, without changing anything, how many target habits they complete each day and how stressed they felt on a one-to-eight rating. That is the baseline phase. For the next two weeks the student follows a structured routine the counsellor designed; that is the intervention phase. For a final two weeks the structured supports are withdrawn but the student continues recording; that is the maintenance phase.

The research question is: **did the student's habit completion and stress ratings change when the routine was introduced, and did any change persist when the supports were removed?**

The case is the student. There is one of them. The forty-two rows are forty-two occasions on which that one student was measured, and consecutive days are related: a good day tends to follow a good day.

Pause and Decide

Before reading further, name the case, state how many independent units the design contains, decide what you would look at first if you could see only one thing, and write the strongest claim an A-B design with a maintenance phase could support. Then say what a very large statistical result would and would not add.

Why This Model?

The primary analysis is visual and descriptive: a chronological plot of each series with the phase changes marked, supported by phase-level summaries of level, trend, and variability, and by a nonoverlap statistic that quantifies how completely the phases separate.

That is the model because the question is about a pattern in one series over time. The interesting features - whether the change was immediate, whether it held, whether the baseline was already moving - live in the sequence, and every one of them is destroyed by summarising each phase as a single mean.

Nonoverlap of all pairs (NAP) is the quantitative summary used here. It takes every possible pairing of one baseline observation with one comparison-phase observation, scores a pair 1 when the comparison observation is better, 0 when it is worse, and 0.5 when the two are tied, and reports the average. It ranges from 0 to 1, with 0.5 meaning complete overlap and 1 meaning that every comparison-phase observation improved on every baseline observation. It is a statement about pairs of days, not about how many habits changed.

The design is an A-B sequence with a follow-up phase. It is the weakest of the standard single-case structures because the change is introduced once and never withdrawn or replicated. Anything that happened at the same

time as the routine - a change in term workload, weather, sleep, a resolved worry - remains an alternative explanation for the whole pattern.

What Would a Favorable Result Look Like?

The counsellor would hope for an immediate change at the phase boundary, little overlap between phases, a stable or contrary baseline trend, and maintenance values that hold rather than drift back. Direction and pattern matter far more than any single summary number.

An unfavourable or mixed pattern is also informative. A gradual change with a rising baseline would suggest the routine did not add much to something already under way. A change that reverses in maintenance would say the effect depends on the supports remaining in place, which is itself a finding a counsellor can act on.

Best Defensible Claim

Statistical evidence: The series describe the level, trend, variability, and overlap of one student's habit counts and stress ratings across three phases, and the nonoverlap statistics quantify how completely the phases separate.

Causal identification: Weak. The routine was introduced once, at a known time, with no withdrawal and no replication, so any concurrent change remains an alternative explanation.

Claim ceiling: The counsellor may describe what happened for this student across these six weeks and note that the change coincided with the routine's introduction. The counsellor may not claim the routine caused the change with any confidence, and may not claim anything at all about another student.

What Must Be True?

Measurement must be comparable across all forty-two days: the same definition of a completed habit and the same stress scale, recorded with the same diligence in every phase. Reactivity is a real concern, because the act of recording can itself change behaviour, and it operates from day one. The phase boundaries must be the dates that were planned, not dates chosen afterwards because the series changed there.

Defensibility Procedures

Plot both series before computing anything. Record the phase dates and the decision rule that set them in advance. Report level, trend, and variability for each phase, and report the nonoverlap statistic with its tie rule and its direction convention stated. Describe the ordinal stress outcome with frequencies and medians, and mark any mean as supplemental. Report both outcomes whether or not they agree.

Next evidence priority: Replicate the same phase structure across several cases with staggered start dates - a multiple-baseline design - so that a change occurring at each case's own phase boundary becomes difficult to explain by a single shared event.

Claim Audit

Repair this claim before continuing: "The routine increased habit completion by 2.43 habits per day and works for students with high stress." Your repair should address both the causal language and the population claim, and should say what the 2.43 figure legitimately describes.

Plausible Research Scenario: The Analyst Who Counted Days as People

This is a fictional research scenario created for teaching. Every referenced dataset is synthetic; it does not describe a real study or real participants.

An analyst opens the same file, sees a grouping column with two values in the first twenty-eight rows and a numeric outcome, and runs an independent-samples t test comparing baseline days with intervention days. The software produces output without complaint.

The habit comparison gives $t(26) = 6.02$, $p < .001$, with a standardized mean difference of 2.28. The stress comparison gives $t(26) = -8.05$, $p < .001$, with a standardized mean difference of -3.04 .

The analyst reports these as the study's inferential results.

Pause and Decide

Before reading further, decide what the number 26 in those degrees of freedom is counting, and whether that count corresponds to anything the study actually collected. Then decide whether the very small p values make the conclusion stronger or make the error more dangerous.

Why This Model?

It is not the model. It is the model to reject, and it is included because it is easy to run, produces impressive output, and is the single most common mistake made with single-case data.

An independent-samples t test assumes that every observation is an independent unit drawn from its group. Here the twenty-eight observations are twenty-eight days in the life of one person. They are not independent of one another in two distinct ways: consecutive days are serially related, and all twenty-eight of them come from one individual whose general tendencies are common to every row. The degrees of freedom of 26 asserts that the study contains 28 independent units. It contains one.

The consequence is not that the estimate is meaningless - the mean difference of 2.43 habits per day is a real description of the two phases - but that the standard error, the t statistic, and the p value are computed from a variability estimate that does not correspond to any replication the study could perform. A p value describes how often a result of this size would arise under repeated sampling; here there is nothing to sample.

What Would a Favorable Result Look Like?

None. A more extreme p value from this analysis is worse, not better, because it makes an unjustified inference harder for a reader to resist.

Best Defensible Claim

Statistical evidence: The mean difference between phases is a descriptive summary of one person's series. The test statistic and p value do not describe any inference the design supports.

Causal identification: Unchanged by the test, and still weak.

Claim ceiling: Descriptive only. No inference to a population and no probability statement.

What Must Be True?

For the t test to be interpretable, days would have to be independent replications, which they are not by construction.

Defensibility Procedures

Report the descriptive difference and the nonoverlap statistic. Do not report a between-groups test on days. If a formal inferential treatment is genuinely needed, use a method built for serially dependent single-case data and state its assumptions - and note that with one case the generalization boundary is unchanged whatever method is used.

Next evidence priority: Additional cases. Statistical machinery cannot substitute for the replication that single-case designs get from repeating the pattern in new participants.

Claim Audit

Explain why " $p < .001$, so the effect is highly reliable" is exactly backwards as a defence of Study 11A.

Problems

Do not consult the solutions until you have recorded an answer, reason, and confidence rating for every attempted item.

Level 1: Recall and Recognize

Problem 15.1 — Count the units

Study 11 contains 42 rows. State how many cases it contains, how many observations, and how many independent units are available for generalizing to other students. Explain why the three answers differ.

Problem 15.2 — Name the design and its weakest feature

The phases run baseline, intervention, maintenance, with the routine introduced once and never withdrawn. Name the design, and name the feature it lacks that stronger single-case designs use to rule out concurrent explanations.

Problem 15.3 — State what NAP measures

Complete the sentence precisely: “Nonoverlap of all pairs reports the proportion of ...” Then state its value under complete overlap and its value when every comparison observation improves on every baseline observation.

Problem 15.4 — Identify the permitted summaries

stress_rating is recorded as an ordinal daily rating from 1 to 8. State which summaries are directly supported by that measurement level, which require an additional assumption, and name the assumption.

Level 2: Predict Before Clicking

Problem 15.5 — Predict overlap from the extremes

In the habit series, the largest baseline value is 4 and the smallest intervention value is 4. Predict, before computing anything, whether any intervention day is worse than any baseline day, and predict roughly what NAP will be.

Problem 15.6 — Predict the effect of the tie rule

Baseline contains two days with a habit count of 4 and intervention contains eight such days. Predict how many of the 196 baseline-intervention pairs are ties, and predict whether the half-credit rule will raise or lower NAP relative to a rule that ignored ties.

Problem 15.7 — Predict what a declining baseline does

The baseline slope is -0.033 habits per day for the habit count and -0.062 rating points per day for stress. For each outcome, predict whether a declining baseline strengthens or weakens the claim that the routine produced the change, and explain the asymmetry.

Problem 15.8 — Predict what the wrong test will show

The two phase means for habit count are 2.357 and 4.786, with standard deviations near 1.0 and 1.1 and 14 observations each. Predict the approximate t statistic from an independent-samples test on days, and predict whether its p value will be small. Then state whether a small value would support the study’s conclusion.

Level 3: Calculate and Explain

Problem 15.9 — Compute a level change

The baseline habit mean is 2.357 and the intervention habit mean is 4.786; the baseline stress mean is 6.571 and the intervention stress mean is 3.786. Compute both level changes with their signs, and state what the sign means for each outcome given the direction of each scale.

Problem 15.10 — Compute NAP by hand

For the habit count, all 196 baseline-intervention pairs fall into three categories: 180 in which the intervention day is higher, 16 ties, and none in which the baseline day is higher. Compute NAP using the half-credit tie rule and show the arithmetic.

Problem 15.11 — Compute NAP for the ordinal outcome

For stress, lower is better. Of the 196 baseline-intervention pairs, 189 show improvement, 7 are ties, and none show deterioration. Compute the improvement NAP and explain why the direction convention has to be stated explicitly for this outcome but not for the habit count.

Problem 15.12 — Convert NAP into a statement about days

The baseline-to-maintenance NAP for stress is 1.000. State exactly what that value means in terms of days, and state what it does not mean about the size of the change.

Level 4: Read jamovi Output**Problem 15.13 — Read the phase descriptives**

Outcome	Phase	<i>n</i>	Median	Q1	Q3	Min	Max	Slope per day	First	Last
Habit count	Baseline	14	2.0	2.0	3.0	1	4	−0.033	3	1
Habit count	Intervention	14	4.0	4.0	5.0	4	7	+0.068	5	6
Habit count	Maintenance	14	5.0	5.0	6.0	3	9	−0.095	5	5
Stress rating	Baseline	14	6.5	6.0	7.0	6	8	−0.062	7	6
Stress rating	Intervention	14	4.0	3.0	4.0	2	6	−0.160	4	3
Stress rating	Maintenance	14	3.0	2.0	3.0	1	4	+0.077	2	3

Describe the level and trend of each series across the three phases. Name the one phase whose range is noticeably wider than the others and say what that widening might reflect.

Problem 15.14 — Read the ordinal frequencies

Stress rating	1	2	3	4	5	6	7	8
Baseline	0	0	0	0	0	7	6	1
Intervention	0	2	3	6	2	1	0	0
Maintenance	1	4	6	3	0	0	0	0

Describe what this table shows that a table of phase means would not. Identify the single value at which the baseline and intervention distributions touch, and say how many days sit there in each phase.

Problem 15.15 — Read the nonoverlap table

Contrast	Outcome	NAP
Baseline to intervention	Habit count	0.959
Baseline to maintenance	Habit count	0.974
Baseline to intervention	Stress (improvement)	0.982
Baseline to maintenance	Stress (improvement)	1.000
Intervention to maintenance	Habit count	0.684
Intervention to maintenance	Stress (improvement)	0.750

Write two sentences describing what this table shows about the introduction of the routine and about what happened after the supports were withdrawn.

Problem 15.16 — Read the output that should not have been produced

Study 11A reports $t(26) = 6.02$, $p < .001$, $d = 2.28$ for the habit count, and $t(26) = -8.05$, $p < .001$, $d = -3.04$ for stress. State what the number 26 asserts about the study, state whether that assertion is true, and identify the one quantity in this output that remains a defensible description.

Level 5: Choose the Model

Problem 15.17 — Choose the primary analysis

Choose the primary analysis for the stated question and justify it from the data structure. State what you would present first, what you would present as quantitative support, and what you would refuse to present at all.

Problem 15.18 — Reject the between-groups test

Reject the Study 11A analysis. Name the assumption it violates, name two distinct sources of dependence in the data, and explain why a more extreme p value makes the error more serious rather than less.

Problem 15.19 — Reject the transition-day claim

An analyst writes: “The effect was immediate: habit count went from 1 on the last baseline day to 5 on the first intervention day, a jump of four.” Reject the inference from those two days, using the phase distributions. Then state what evidence in the series does legitimately support an immediacy claim.

Problem 15.20 — Choose the design change that would help most

Rank these three changes by how much each would strengthen the causal claim, and justify the ranking: extending each phase from 14 days to 28 days; adding a return-to-baseline phase after the intervention; running the same phase structure in four more students with staggered start dates.

Level 6: Report and Critique

Problem 15.21 — Write the result paragraph

Using the verified values, write a results paragraph for Study 11. Include the case and phase structure, the level and trend of both series, the ordinal frequencies for stress, the nonoverlap statistics with the tie rule, and the design limitation.

Problem 15.22 — Repair a generalization claim

Repair this sentence: “The structured routine raises habit completion and lowers stress, with NAP values of .96 and .98.” Preserve everything the study does support.

Problem 15.23 — Repair an ordinal-scale report

Repair this sentence: “Mean stress fell from 6.57 to 3.79, a decrease of 2.79 points on the stress scale.” The arithmetic is correct. Identify what must be added or changed for the sentence to be defensible.

Problem 15.24 — Write the limitation

Write the four-sentence limitation for Study 11. One sentence must address the number of cases, one the single unreplicated phase change, one measurement reactivity, and one the maintenance phase specifically.

Solutions

Level 1 Solutions

Solution 15.1

One case, 42 observations, and one independent unit for generalization.

The three answers differ because they count different things. The 42 rows are occasions of measurement, and they support a detailed description of how one person's behaviour moved over six weeks. The case is the entity whose behaviour is being described, and there is one. Generalization to other students requires variation across students, and this study contains none, so no number of additional days would create it. Decisive cue: ask what could be replicated. Another day is a repetition within the same case; another student is a replication of the case.

Solution 15.2

It is an A-B design with a follow-up or maintenance phase - baseline, intervention, then withdrawal of supports with continued measurement.

What it lacks is replication of the phase change. Stronger single-case designs repeat the introduction and removal of the condition within the same case (a reversal or A-B-A-B design) or stagger the introduction across several cases or settings (a multiple-baseline design). Both work the same way: they make a coincidence explanation harder, because a concurrent event would have to align with each phase change independently. With one change at one time, any event that happened in that same week explains the whole pattern equally well.

Solution 15.3

"Nonoverlap of all pairs reports the proportion of all baseline-by-comparison-phase pairs of observations in which the comparison observation is better than the baseline observation, with tied pairs counted as half." Under complete overlap, when the two phases are indistinguishable, it takes the value 0.5. When every comparison observation improves on every baseline observation, it takes the value 1.

Solution 15.4

An ordinal rating directly supports frequencies, the median, quartiles, the range, and any rank-based comparison, because all of these depend only on the ordering of the values. A mean, a standard deviation, and a mean difference require the additional assumption that the intervals between consecutive rating points are equal - that the step from 6 to 7 represents the same amount of stress as the step from 2 to 3. That assumption is not established by the scale and should be named whenever such a summary is reported. The verified record for this study marks its stress means and standard deviations as supplemental for exactly this reason.

Level 2 Solutions

Solution 15.5

No intervention day is worse than any baseline day. The largest baseline value equals the smallest intervention value, so the two distributions touch at 4 and nowhere else, and every remaining pair favours the intervention day. NAP will therefore be very close to 1 - short of it only because the pairs at the point of contact are ties. The verified value is 0.959. Transfer cue: comparing the extremes of two phases tells you the shape of the overlap before any statistic is computed.

Solution 15.6

Ties occur only where both phases contain the same value, which here is only at 4. With two baseline days and eight intervention days at that value, there are $2 \times 8 = 16$ tied pairs out of 196.

The half-credit rule lowers NAP relative to a rule that discarded ties and computed the proportion among the remaining pairs, because those 16 pairs would then vanish and the remaining 180 would all favour the

intervention, giving 1.000. Half credit treats a tie as evidence of neither improvement nor deterioration, which is the more honest treatment and the reason the rule must be stated when NAP is reported.

Solution 15.7

For the habit count, a declining baseline strengthens the claim. Habit completion was drifting slightly downward before the routine began, so an increase afterwards runs against the pre-existing direction and cannot be explained by simple continuation of the baseline trend.

For stress, a declining baseline weakens the claim, because stress was already falling. Some portion of the subsequent decline could be a continuation of what was already under way rather than an effect of the routine.

The asymmetry exists because a baseline trend is only a threat when it points in the same direction as the hoped-for change. The magnitudes matter too: over 14 days the baseline slopes imply drifts of about -0.43 habits and -0.80 rating points across the phase, both small relative to the observed level changes of $+2.43$ and -2.79 , so the stress concern is real but modest.

Solution 15.8

With a mean difference of about 2.43, standard deviations near 1.0 and 1.1, and 14 observations per phase, the pooled standard error is roughly 0.40, giving a t near 6. The verified value is 6.02 with $p < .001$.

A small p value does not support the study's conclusion. It is produced by treating 28 days as 28 independent units, and its smallness is a direct consequence of that error: the more days are recorded, the more independent replications the test believes it has, and the smaller the p value becomes - without a single additional student having been observed.

Level 3 Solutions

Solution 15.9

Habit count: $4.786 - 2.357 = +2.43$ habits per day. Stress: $3.786 - 6.571 = -2.79$ rating points.

The signs mean opposite things because the scales run in opposite directions. Higher habit counts are the desired outcome, so a positive level change is an improvement. Higher stress ratings are the undesired outcome, so a negative level change is an improvement. Both series therefore moved in the favourable direction, and any statistic reported for stress must have its direction convention stated or a reader cannot tell which sign is good news.

Solution 15.10

$$\text{NAP} = (180 \times 1 + 16 \times 0.5 + 0 \times 0) / 196 = (180 + 8) / 196 = 188 / 196 = 0.959.$$

Every one of the 196 pairs is counted exactly once, so the denominator is the total number of baseline-by-intervention pairings, 14×14 . The 16 ties contribute 8, which is what pulls the value below 1.

Solution 15.11

$$\text{NAP} = (189 \times 1 + 7 \times 0.5) / 196 = (189 + 3.5) / 196 = 192.5 / 196 = 0.982.$$

The direction convention has to be stated for stress because improvement means a *lower* value. Without that statement, a reader cannot tell whether 0.982 means that intervention days were mostly lower - which is the intended reading - or mostly higher. For the habit count the desirable direction is upward and coincides with the numerical ordering, so the convention is less likely to be misread, but stating it there too costs nothing and prevents an error when the two outcomes are reported side by side.

Solution 15.12

A baseline-to-maintenance NAP of 1.000 for stress means that every one of the 196 pairings of a baseline day with a maintenance day showed the maintenance day at a lower stress rating, with no ties. The two phases do not overlap at all: the lowest baseline rating, 6, is above the highest maintenance rating, 4.

It does not mean the change was large. NAP measures separation, not magnitude. Two phases could separate completely with a difference of one rating point, and they would earn the same 1.000. The magnitude comes from the medians, 6.5 against 3.0, and the frequency table, and it must be reported alongside the nonoverlap statistic rather than inferred from it.

Level 4 Solutions

Solution 15.13

The habit series rises across the three phases: median 2.0 in baseline, 4.0 in intervention, 5.0 in maintenance. Baseline drifts slightly downward (slope -0.033) while intervention drifts slightly upward ($+0.068$), so the change at the first phase boundary runs against the pre-existing direction. Maintenance holds the higher level with a slight downward slope (-0.095).

The stress series falls across the three phases: median 6.5, then 4.0, then 3.0. Baseline is already declining slightly (-0.062), intervention declines more steeply (-0.160), and maintenance is flat to slightly rising ($+0.077$), which is consistent with the level settling rather than continuing to fall.

The maintenance habit phase has the widest range, from 3 to 9 against baseline's 1 to 4 and intervention's 4 to 7. That widening could reflect genuinely more variable behaviour once the external structure was removed, a single unusually productive day, or ordinary variation in a 14-day window. A range is driven by two observations, so the phase's quartiles - 5.0 and 6.0, an interquartile range of 1, the same as the other two phases - are the better guide to whether the middle of the distribution has actually spread out. Here it has not.

Solution 15.14

The frequency table shows the *shape* of each phase's distribution, not only its centre. Baseline stress occupied only three values, 6, 7, and 8, with no day below 6. Intervention spread across 2 to 6 with a mode at 4. Maintenance occupied 1 to 4 with a mode at 3. Phase means would show three numbers falling; the frequency table additionally shows that the phases are almost disjoint, that baseline had no good days at all rather than a mixture, and that variability changed shape rather than merely shrinking.

Baseline and intervention touch at the single value 6: seven baseline days and one intervention day sit there, which accounts for the 7 tied pairs in the nonoverlap calculation.

Study in the Larger Evidence Program: One Student's Routine

- **What this study adds:** A detailed, reproducible description of how one student's habit completion and stress ratings moved across the introduction and withdrawal of a structured routine, with near-complete phase separation on both outcomes.
- **What it cannot settle:** Whether the routine caused the change for this student, and anything at all about any other student.
- **Causal-support role:** Weak responsiveness evidence. Something was deliberately introduced at a known time, which is more than an observational study offers, but the change was not replicated or withdrawn, so a concurrent event explains the pattern equally well.
- **Next evidence priority:** A multiple-baseline design across four or five students with staggered start dates, so that the change must coincide with each case's own phase boundary rather than with one calendar week.

Solution 15.15

Introducing the routine coincided with near-complete separation on both outcomes: nonoverlap of all pairs was .96 for habit count and .98 for stress improvement, with no baseline-intervention pair favouring the baseline day in either series and the shortfall from 1.00 due entirely to tied pairs. Separation was as strong or stronger against the maintenance phase (.974 and 1.000).

After the supports were withdrawn, neither outcome returned toward baseline; both continued to improve modestly, with intervention-to-maintenance NAP values of 0.684 for habit count and 0.750 for stress. Those values are well above 0.5, indicating continued gain, but far below the baseline contrasts, indicating that the two later phases substantially overlap - the maintenance phase is better than the intervention phase on most pairs of days, not on nearly all of them.

Solution 15.16

The 26 asserts that the analysis has 28 independent observations from which two group means were estimated. That assertion is not true. The study contains one case measured on 28 occasions; the days are serially related to one another and all share the characteristics of a single person.

The one quantity that remains defensible is the mean difference between phases - 2.43 habits per day and -2.79 rating points - as a *description* of what the two phases looked like for this student. The standardized effect sizes of 2.28 and -3.04 are borderline: they are computed from a within-phase standard deviation that describes day-to-day variability in one person, so they should not be read against benchmarks built for between-person comparisons. The *t* statistics, the degrees of freedom, and the *p* values describe an inference the design cannot support and should not be reported at all.

Level 5 Solutions

Solution 15.17

The primary analysis is a chronological plot of each series with the phase boundaries marked, read for level, trend, variability, and immediacy of change. That is what should be presented first, because every feature that distinguishes a convincing single-case result from an unconvincing one is visible in the sequence and invisible in any summary.

The quantitative support is the set of phase-level summaries - median, quartiles, range, and slope for each phase - together with the nonoverlap statistics with their tie rule and direction conventions stated, and the ordinal frequency table for stress.

What should be refused entirely is any between-groups inferential test on days, and any *p* value or confidence interval presented as if the 42 observations were independent units. Descriptive means for the ordinal stress outcome may be reported but must be marked supplemental with the equal-interval assumption named.

Solution 15.18

The violated assumption is independence of observations within groups, which an independent-samples *t* test requires in order for the standard error to describe the variability of the estimate under replication.

Two distinct sources of dependence are present. The first is serial dependence: consecutive days in one person's life are related, so day 8 carries information about day 9, and the effective amount of independent information is smaller than the number of rows. The second is the single case itself: all 28 observations come from one person, so everything stable about that person - their baseline capacity, their scale use, their circumstances that week - is common to every row and is never sampled.

A more extreme *p* value makes the error more serious because the size of the error grows with the number of observations. Recording the same student for 84 days instead of 28 would roughly triple the apparent degrees of freedom and shrink the *p* value further, while adding no new independent unit whatsoever. A result that becomes more impressive as the underlying justification stays constant is a warning, not a confirmation.

Solution 15.19

The last baseline day happened to be 1, the minimum value of the entire baseline phase, and the first intervention day was 5, which sits above the intervention median of 4.0, at that phase's third quartile. The apparent jump of four is therefore a comparison of an unusually low day with a fairly typical one, and it exaggerates the change. Comparing the phase medians of 2.0 and 4.0 gives a difference of two, and the level

change in means is 2.43. Any two adjacent days can differ by three in a series whose baseline already ranges from 1 to 4.

What legitimately supports an immediacy claim is the behaviour of the whole distribution at the boundary rather than the boundary pair: the intervention phase's minimum, 4, equals the baseline phase's maximum, so from the very first intervention day onward every value sits at or above everything seen in the baseline phase, and no intervention day falls into the baseline range. That is a statement about all 14 days, and it does not depend on which day happened to be last.

Solution 15.20

Ranked from most to least helpful.

Running the same phase structure in four more students with staggered start dates helps most. It adds independent units, which is the one thing this study entirely lacks, and staggering makes a concurrent-event explanation implausible because the change would have to arrive at each student's own boundary. It addresses both the causal weakness and the generalization boundary.

Adding a return-to-baseline phase helps second. It replicates the phase change within the case, so a concurrent event would have to arrive and then depart on schedule. It addresses the causal weakness but adds no new case and so does nothing for generalization, and it may be unacceptable if withdrawing a beneficial routine harms the participant.

Extending each phase to 28 days helps least. Longer phases give more stable estimates of level and trend and make the series easier to read, which is worth something. They add no replication of the phase change and no new case, so the causal claim and the generalization boundary are exactly where they were. This is the change most likely to be mistaken for an improvement in evidence because it increases the number of rows.

Study in the Larger Evidence Program: What Replication Would Add

- **What this study adds:** As a single case, it contributes a well-documented instance and a demonstration that phase separation can be near-total on both outcomes.
- **What it cannot settle:** Whether the pattern is attributable to the routine, and whether it would appear in anyone else. Neither gap closes with more days.
- **Causal-support role:** Weak responsiveness evidence from one deliberate, unreplicated introduction.
- **Next evidence priority:** Four or five cases with staggered baselines. If each case's change arrives at that case's own phase boundary, a single concurrent event stops being a credible explanation, and the accumulation of cases begins to address generality.

Level 6 Solutions

Solution 15.21

One student recorded daily habit completion and a daily ordinal stress rating for 42 consecutive days, in three consecutive 14-day phases: baseline, a structured routine, and a maintenance phase in which the supports were withdrawn. Habit count rose across the phases, with medians of 2.0, 4.0, and 5.0 and phase ranges of 1-4, 4-7, and 3-9; the baseline trend was slightly negative (-0.033 per day), so the increase ran against the pre-existing direction. Stress ratings fell, with medians of 6.5, 4.0, and 3.0; baseline ratings occupied only the values 6 to 8, intervention ratings 2 to 6, and maintenance ratings 1 to 4. Nonoverlap of all pairs, with tied pairs credited 0.5, was .96 for habit count and .98 for stress improvement from baseline to intervention, and .97 and 1.00 respectively from baseline to maintenance; intervention-to-maintenance values were .68 and .75. Means and standard deviations for the ordinal stress rating are reported as supplemental summaries and assume equal spacing between adjacent rating points. Because the study involves one case and one unreplicated phase change, these results describe what happened for this student and do not establish that the routine caused the change or that it would produce a similar pattern in anyone else.

Solution 15.22

Repaired: “For this student, habit completion rose and stress ratings fell when the structured routine was introduced, with nonoverlap of all pairs of .96 and .98 against the baseline phase. Because the routine was introduced once in a single case, with no withdrawal and no replication in other students, the design cannot establish that the routine produced the change, and the result says nothing about how another student would respond.”

The original sentence makes two jumps at once. The present tense - “raises,” “lowers” - converts one person’s series into a general property of the routine, and the absence of any subject makes the claim about students in general. Both are unsupported by a single unreplicated case, however clean its numbers.

Solution 15.23

The arithmetic is right: $6.571 - 3.786 = 2.79$. Two things must change.

First, the sentence treats an ordinal rating as an interval scale. A mean of an ordinal variable, and a difference between two such means, require the assumption that the distance from 6 to 7 equals the distance from 2 to 3. That assumption is not established by a self-report rating scale and must be stated when the summary is used.

Second, better summaries are available and should lead. The medians, 6.5 and 4.0, and the frequency distributions convey the change without any additional assumption, and the frequencies convey more: baseline occupied only 6, 7, and 8, while intervention spread from 2 to 6.

Defensible version: “Stress ratings fell from a baseline median of 6.5, with all 14 days between 6 and 8, to an intervention median of 4.0, with days spread from 2 to 6. The corresponding means are 6.57 and 3.79; these are reported as supplemental summaries and assume equal spacing between adjacent points of the ordinal rating scale.”

Solution 15.24

“The study describes one case, so it provides no basis for expecting a similar pattern in another student, and no increase in the number of recorded days would change that. The routine was introduced once and never withdrawn or reintroduced, so any event coinciding with that single phase boundary - a change in workload, sleep, health, or circumstances during that week - explains the whole pattern as well as the routine does. Because the student recorded both outcomes daily throughout, the act of measurement may itself have influenced the behaviour being measured, and that influence was present in every phase including the baseline. Finally, the maintenance phase shows that the improved level held for two weeks after the supports were withdrawn, which is a short follow-up window and cannot speak to whether the pattern would persist over a term or a year.”

Cross-Volume Model-Choice Review

The units in this volume are not a list of substitutes for the methods in Volume 1. Each answers a question the earlier methods cannot pose. This table is a review aid for that distinction; it is not a flowchart, and reading down the right-hand column is not a substitute for reading a design.

The question actually being asked	What one row represents	Defensible analysis	Where it is treated
How much did a quantity change within the same people?	One person measured twice	Paired comparison of differences	Volume 1, Unit 10
How far apart are two independent group means?	One person in one group	Independent-samples comparison, variance model chosen from the design	Volume 1, Unit 10
How far apart are three or more independent group means?	One person in one group	One-way analysis of variance with a prespecified multiplicity-adjusted family	Volume 1, Unit 11
Does the distribution of a categorical choice differ across groups?	One person, one category	Chi-square test of independence with an association effect size	Unit 12
Do observations in one group tend to sit higher in the ordering than those in another?	One person, one score	Kruskal-Wallis with jamovi epsilon-squared, a bias-adjusted rank effect size, and adjusted pairwise comparisons	Unit 13
How does an outcome differ across groups that were never assigned?	One person in one observed or unassignable group	Group comparison, reported with the identification limit stated	Unit 14
How much of a group difference does not operate through a third variable?	One person, plus the third variable	Covariate-adjusted model, reported as a decomposition and not a correction	Unit 14
Did one person's behaviour change when a condition was introduced?	One occasion for one case	Series inspection with level, trend, variability, and nonoverlap	Unit 15

Four questions that decide most of this volume

Is the outcome a quantity or a category? If it is a category, no mean exists and no difference can be reported in the outcome's units. Recoding categories as numbers changes the answer when the labels are reordered, which is the test for whether the recoding was legitimate.

Is the reason for a rank method the shape of the distribution or something else? Skew and extreme values are reasons. Unequal spread is not, and a rank method is not a remedy for it. Whatever the reason, state it, and report the result on the scale the method actually estimates.

Was the grouping variable assigned, observed, or unassignable? This determines the claim ceiling and nothing in the analysis can change it. An unassignable grouping variable such as age has a permanently lower ceiling, because there is no counterfactual in which the same case occupies a different level at the same moment.

For every covariate, where does it sit in the causal order? A common cause measured before both variables is a confounder, and adjusting for it reduces bias. A consequence of the grouping variable is not, and adjusting

for it removes part of the effect. The regression output looks identical in both cases; only an argument about the causal order distinguishes them.

Verified Result Records and How to Check Your Output

Every numerical value printed in this volume's worked solutions is recomputed from the source CSV files by a public audit script in the same repository, and every value in the verified result records was independently recomputed from the same files before publication. This section explains how to use those records when your own output disagrees.

What is published

For each of the four studies used here, the repository contains:

- the synthetic CSV file;
- a machine-readable data dictionary naming every variable's role, measurement level, permitted values, and missing-value policy;
- a verified result record in JSON holding the analysis targets for that study;
- SHA-256 checksums for the data files and the result records; and
- an entry in ANALYSIS_MATRIX.md mapping the dataset to its intended workflows and its result fields.

Nothing in that list requires an account, a purchase, or a prepared jamovi session file.

The four causes of nearly every mismatch

When a number on your screen disagrees with a number in this volume, work through these before reporting an error.

The import. Confirm the row count and the column names against the dictionary. Confirm that the file you opened is the one the problem names, and that it was imported fresh rather than restored from a saved session.

The measurement level. jamovi guesses a level on import and its guess is sometimes wrong. A grouping variable read as continuous, or an identifier read as nominal and included in an analysis, changes the output without any warning.

The group order. A difference has a sign, and the sign depends on which group the software subtracts from which. Every direction in this volume is stated explicitly in the problem; check yours against it before concluding that a value is wrong.

Missing values. Blank cells must be treated as missing rather than as zero, and the number of complete cases in your output must match the record.

When the disagreement survives all four

Then it is worth reporting, and it is worth reporting precisely. Include your operating system, your jamovi release, the dataset filename, the exact analysis and options you selected, the observed value, and the expected value. Use the repository's structured issue forms for a reproducible error and Discussions for a question. Do not attach real participant, student, clinical, institutional, thesis, or restricted data to any report.

A note on version identity

The verified records for these studies were produced against jamovi Desktop application release 28.2, whose bundled analysis modules report version 28.2.0. The Flatpak metadata field for the same installation reports 2.7.27, and the project's own requested citation family is Version 2.7. These are three labels for one installation and serve different records; none of them contradicts the others. When you report your own environment, give the release your application displays together with your operating system.

A note on the effect sizes reported after Kruskal-Wallis

jamovi reports epsilon-squared as $H / (N - 1)$. For these data, $39.573 / 131 = 0.302$. Unit 13 uses that value whenever it names jamovi epsilon-squared, matching the software output learners are asked to read.

The unit also reports a separately named bias-adjusted rank effect size, $(H - k + 1) / (N - k) = 0.291$. That formula subtracts the value the statistic would be expected to take under no association before rescaling it, so the adjusted measure is centred at zero when nothing is happening. The two quantities answer closely related questions and the gap between them shrinks as the sample grows, but they are not interchangeable labels. Report the formula or software definition and do not call the 0.291 quantity jamovi epsilon-squared.

The same caution applies to any effect size whose definition varies between packages. Naming the formula alongside the number costs one clause and removes the ambiguity entirely.

A note on pairwise comparisons after Kruskal-Wallis

Unit 13 reports planned Mann-Whitney comparisons with tie-adjusted variance, a continuity correction, and Holm adjustment applied across the family of three. jamovi's Kruskal-Wallis analysis offers its own built-in pairwise comparison procedure, which is a different method and will produce different numbers. To reproduce the values printed in Unit 13, filter to each pair of groups in turn, request the Mann-Whitney U comparison in the independent-samples analysis, and apply the Holm adjustment across the three resulting p values. Either approach is defensible; reporting one while citing the other is not.

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About the Author

Nicholas Elliott Karlson writes open, reproducible materials for teaching research methods and statistics. His work emphasizes model choice, claim limits, synthetic teaching data, independently verified results, and workflows that a reader can inspect from the raw file through the reported conclusion.

About This Resource

Psychology Statistics Practice Materials with Jamovi, Volume 2: Categorical, Rank-Based, Quasi-Experimental, and Single-Case Evidence contains four units, 96 practice problems, and 96 fully worked solutions. It continues the unit numbering of Volume 1 and is licensed under the same Creative Commons Attribution 4.0 International License.

All research scenarios are fictional and all teaching data are synthetic. No real participant, student, patient, client, clinical, institutional, thesis, or restricted data are included, and no result in this resource is evidence about real people.

The synthetic datasets, data dictionaries, verified result records, checksums, and the numerical audit script that recomputes every printed value are published in the same public repository:

<https://github.com/nicholaskarlson/data>

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